Anatomy of the Ear

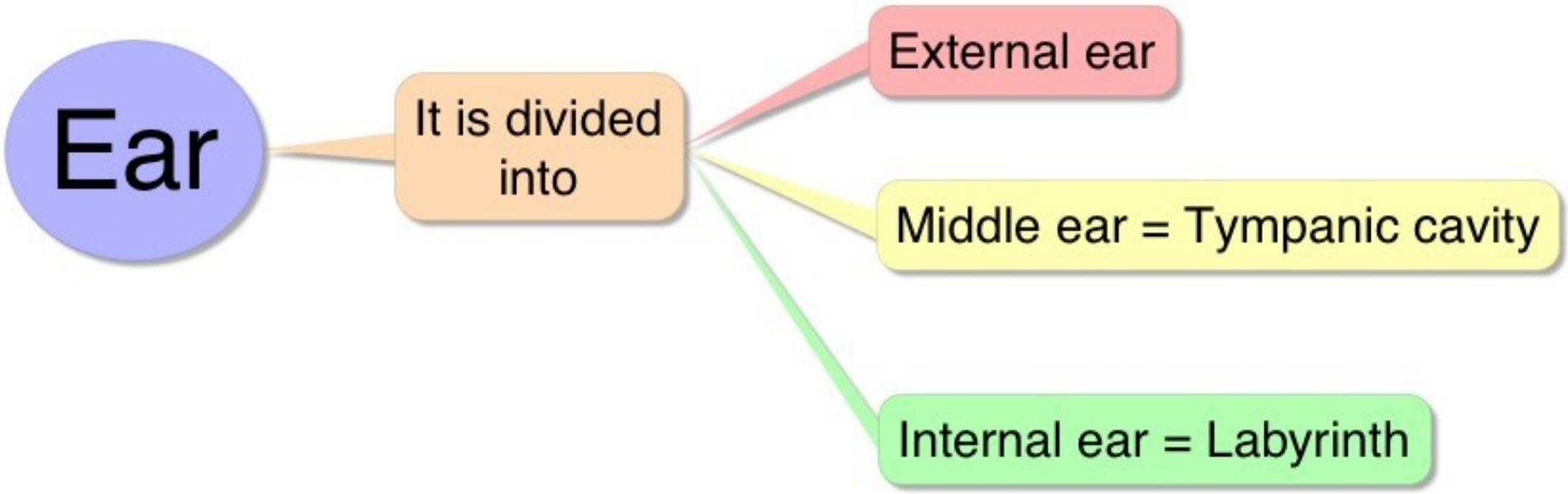
ILOs

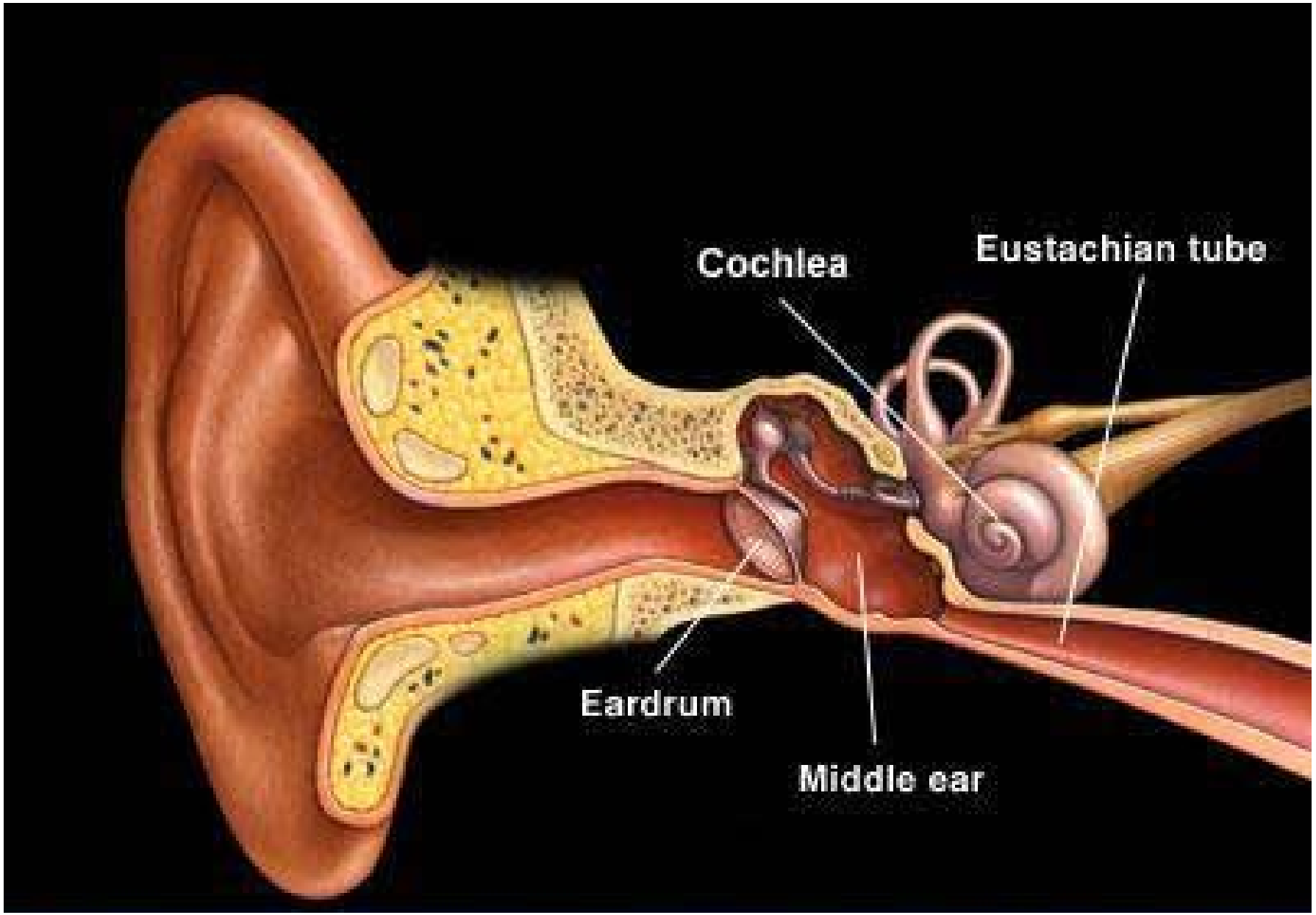
By the end of this lecture, you should be able to:

- List parts of the ear.
- Describe parts of External ear: auricle & external auditory meatus and give their nerve supply and its applied anatomy.
- Describe roof, floor, and walls of Middle ear and enumerate its contents.
- Identify parts and function of auditory tube.
- Name parts of Internal ear: bony & membranous labyrinth.

The Ear

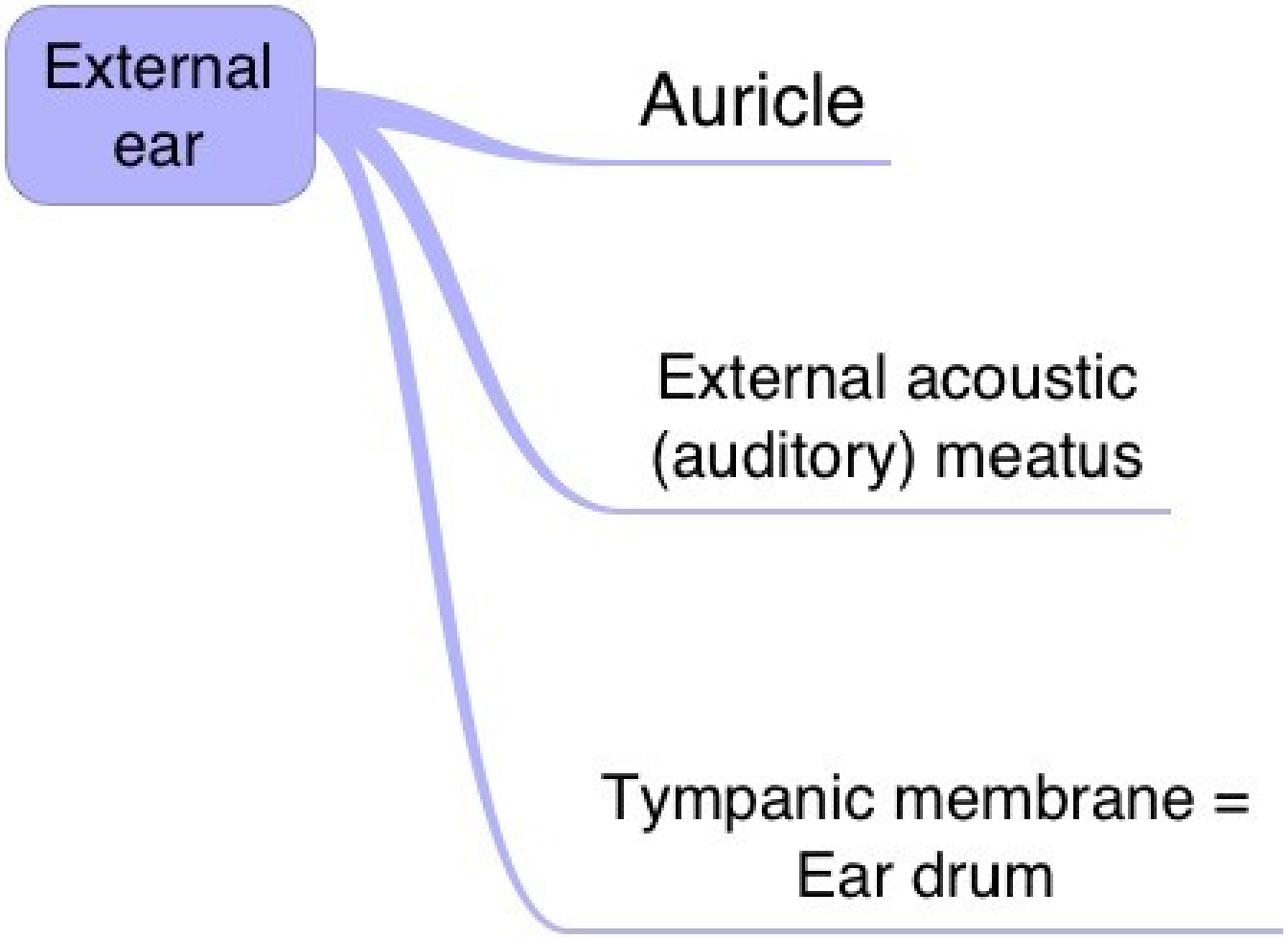
It is the organ of **Hearing**
and
also concerned with **Equilibrium**

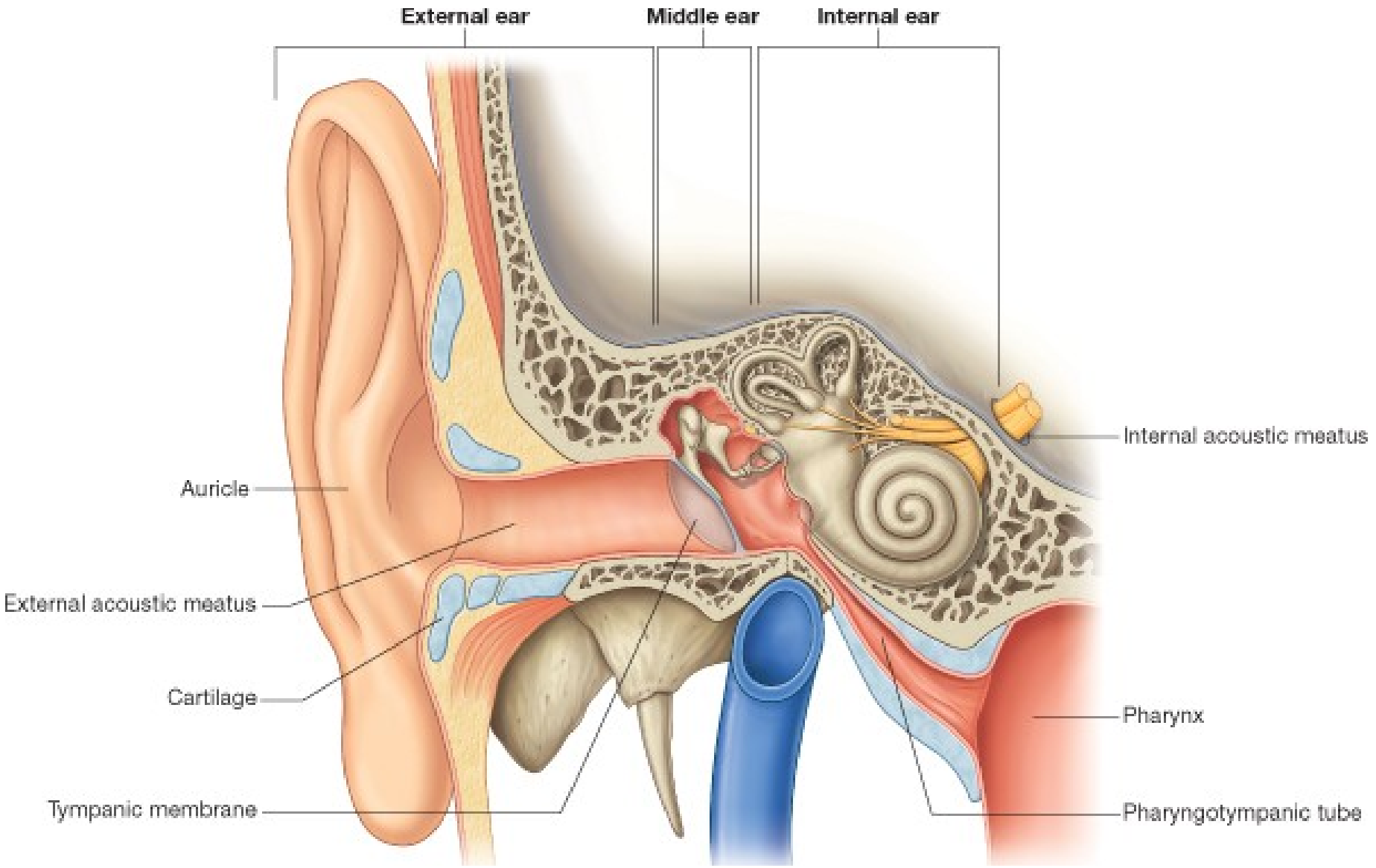

Ear
It is divided into
External ear
Middle ear = Tympanic cavity
Internal ear = Labyrinth


Cochlea
Eustachian tube
Eardrum
Middle ear

External Ear

The external ear can be split functionally and structurally into two sections; the auricle (or pinna), and the external acoustic meatus.


External ear
Auricle
External acoustic (auditory) meatus
Tympanic membrane = Ear drum

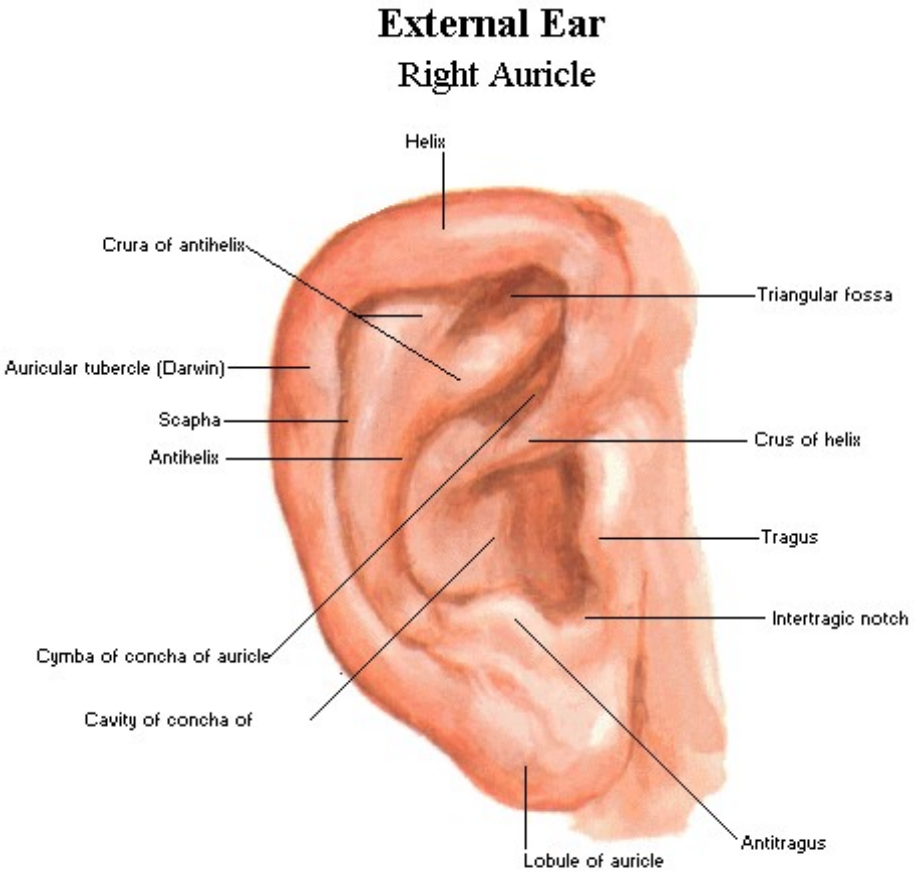

External ear
Middle ear
Internal ear
Internal acoustic meatus
Auricle
External acoustic meatus
Cartilage
Pharynx
Tympanic membrane
Pharyngotympanic tube

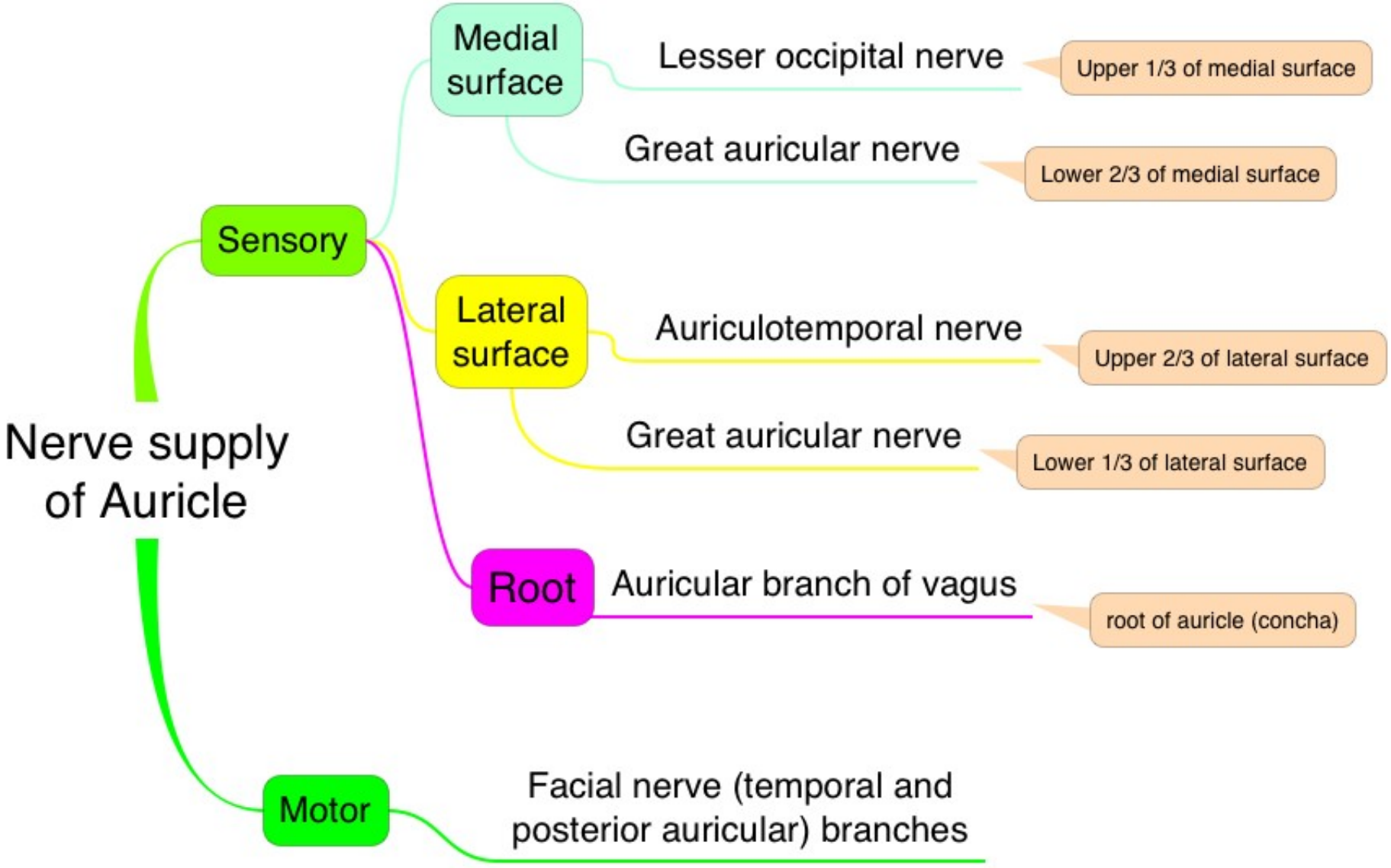
AURICLE

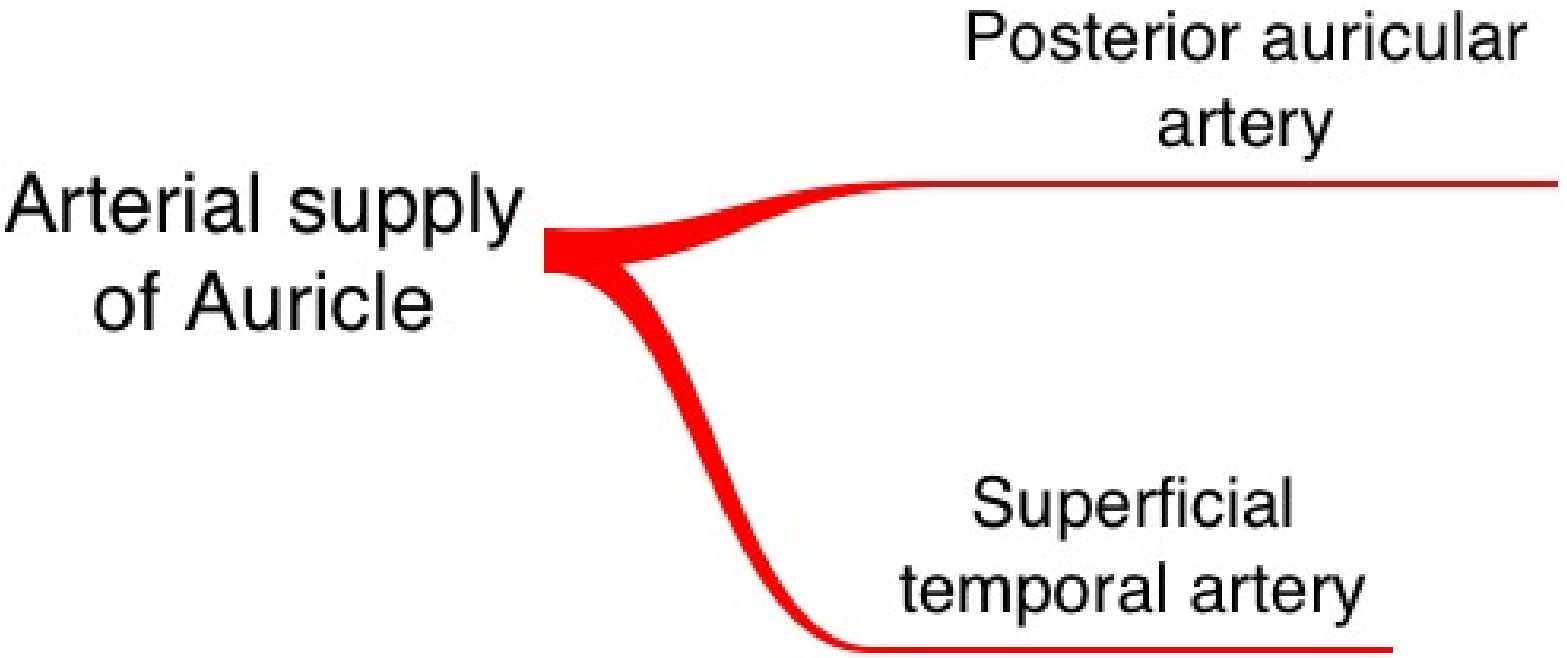
The auricle is an external, lateral paired structure positioned on either side of the head. It can also be known as the pinna. Its function is to **capture** and **transmit** sound to the external acoustic meatus.

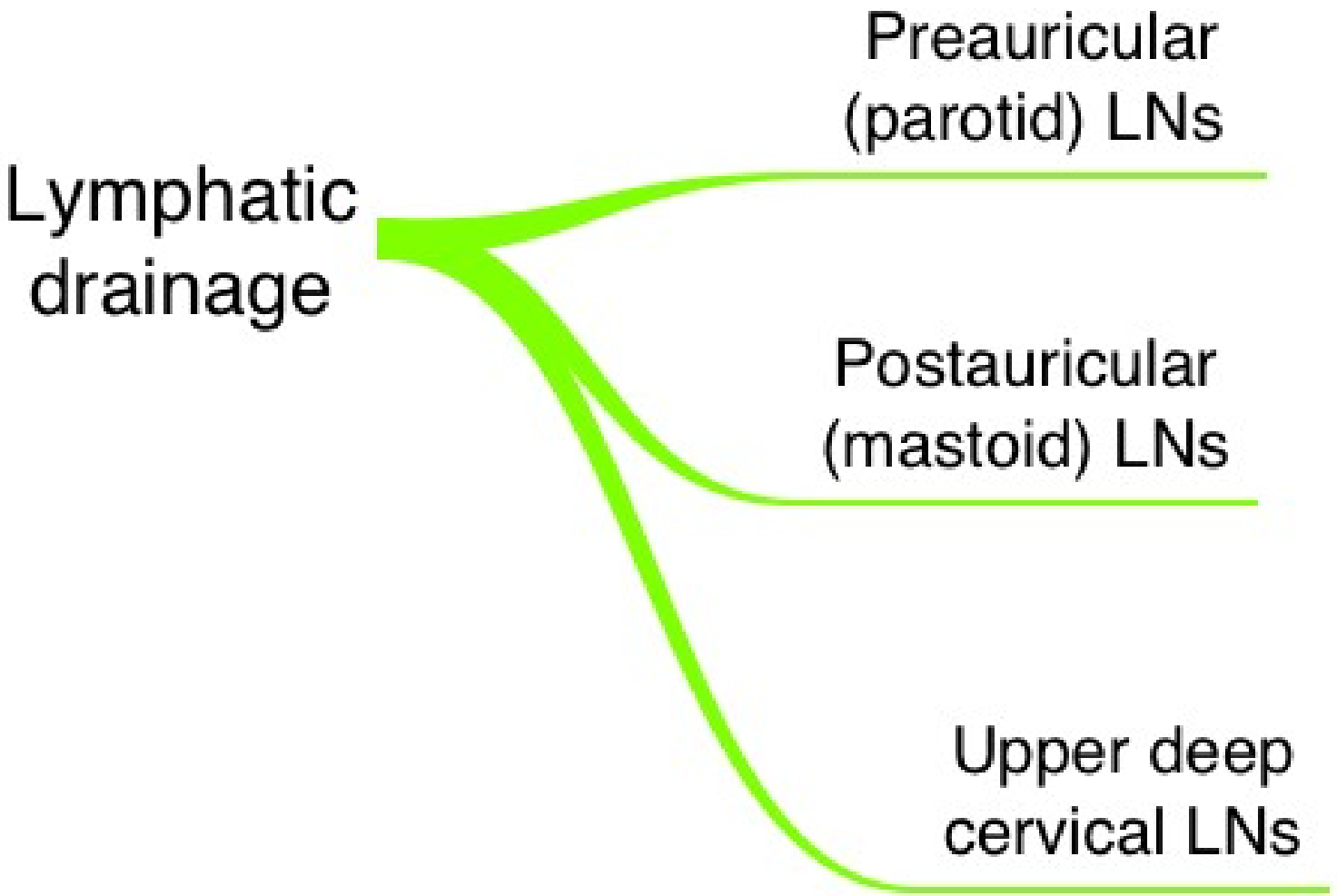
STRUCTURE

Most of the auricle has a cartilaginous framework, with the **lobule** being the only part that is not supported by cartilage. The outer curvature of the ear is called the **helix.** Moving inwards, there is another curved elevation running parallel to the helix – this is known as the **antihelix**. The antihelix divides into two **cura** – the inferoanterior crus, and the superoposterior crus
In the middle of the auricle is a hollow **depression**, called the **concha**. It continues into the skull as the external acoustic meatus. The concha functions to direct sound into the external acoustic meatus. Immediately **anterior** to the start of the external acoustic meatus is an elevation of tissue – the **tragus**. Opposite the tragus is the **antitragus**.




Nerve supply of Auricle
Sensory
Medial surface
Lesser occipital nerve
Upper 1/3 of medial surface
Great auricular nerve
Lower 2/3 of medial surface
Lateral surface
Auriculotemporal nerve
Upper 2/3 of lateral surface
Great auricular nerve
Lower 1/3 of lateral surface
Root
Auricular branch of vagus
root of auricle (concha)
Motor
Facial nerve (temporal and posterior auricular) branches


Arterial supply of Auricle
Posterior auricular artery
Superficial temporal artery

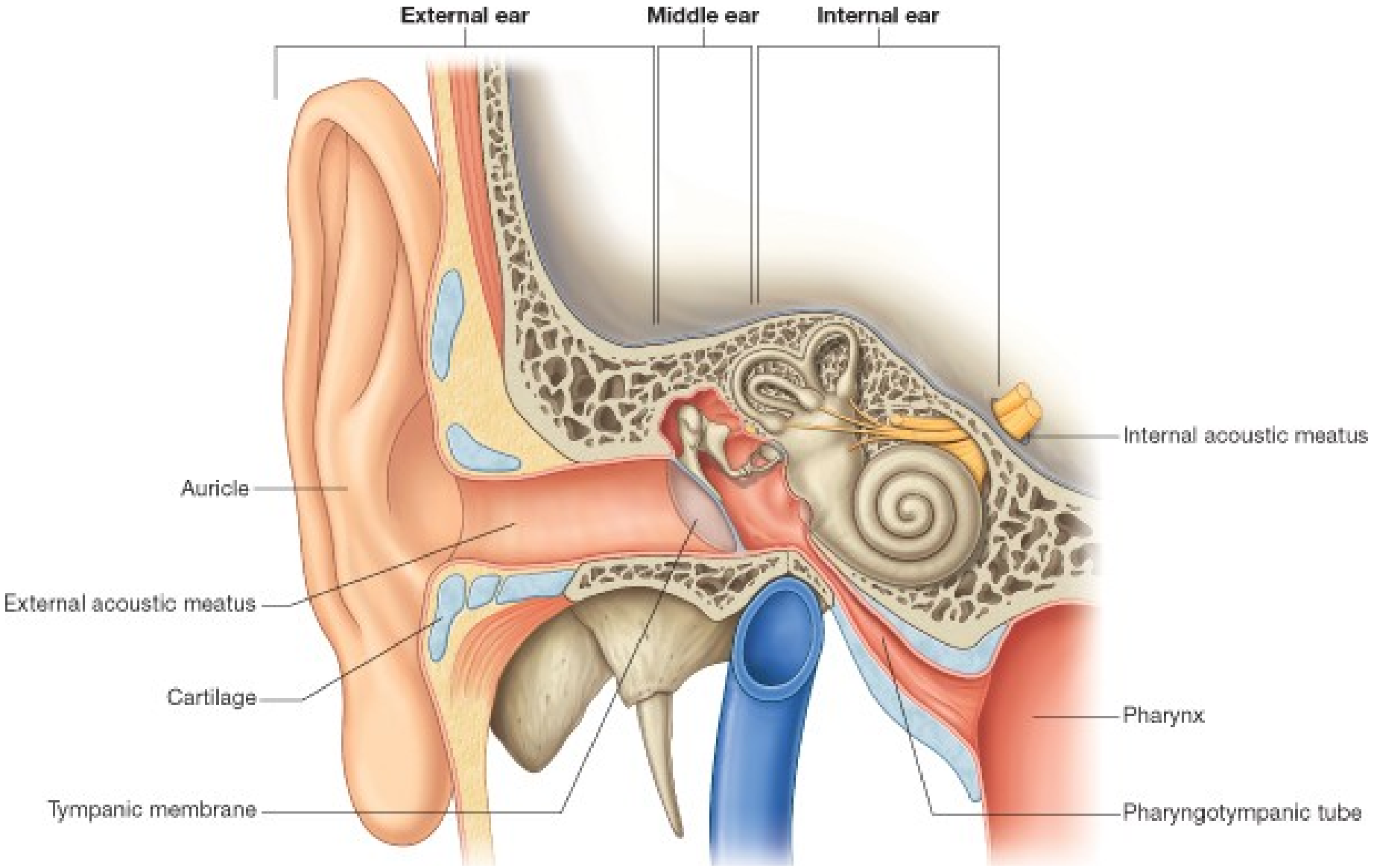

Lymphatic drainage
Preauricular (parotid) LNs
Postauricular (mastoid) LNs
Upper deep cervical LNs

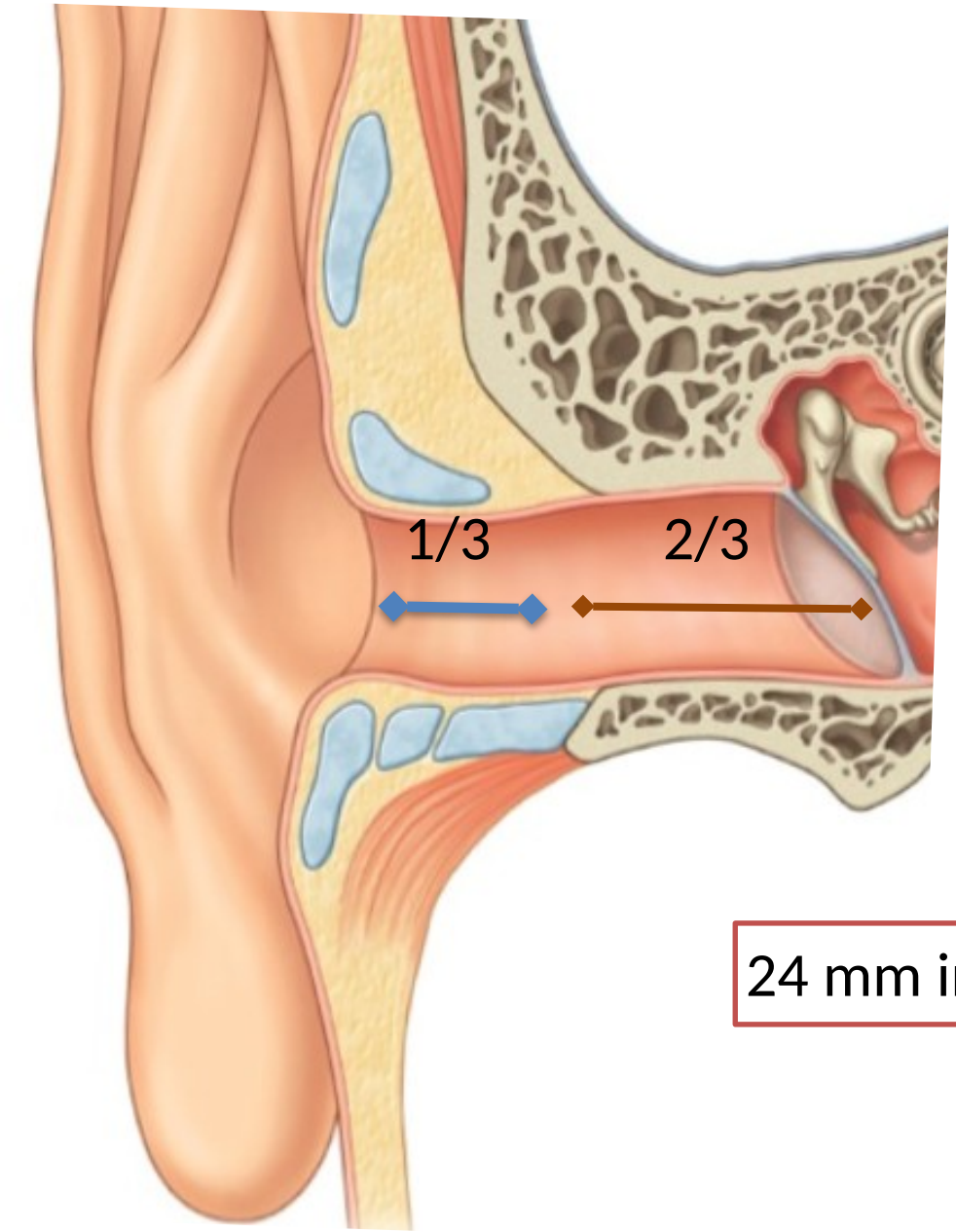
EXTERNAL ACOUSTIC (AUDITORY) MEATUS

The external acoustic meatus is a sigmoid shaped tube that extends from the deep part of the concha to the **tympanic membrane**. The walls are structurally supported by **cartilage** within the auricle, and bony support is provided by the **temporal bone**.

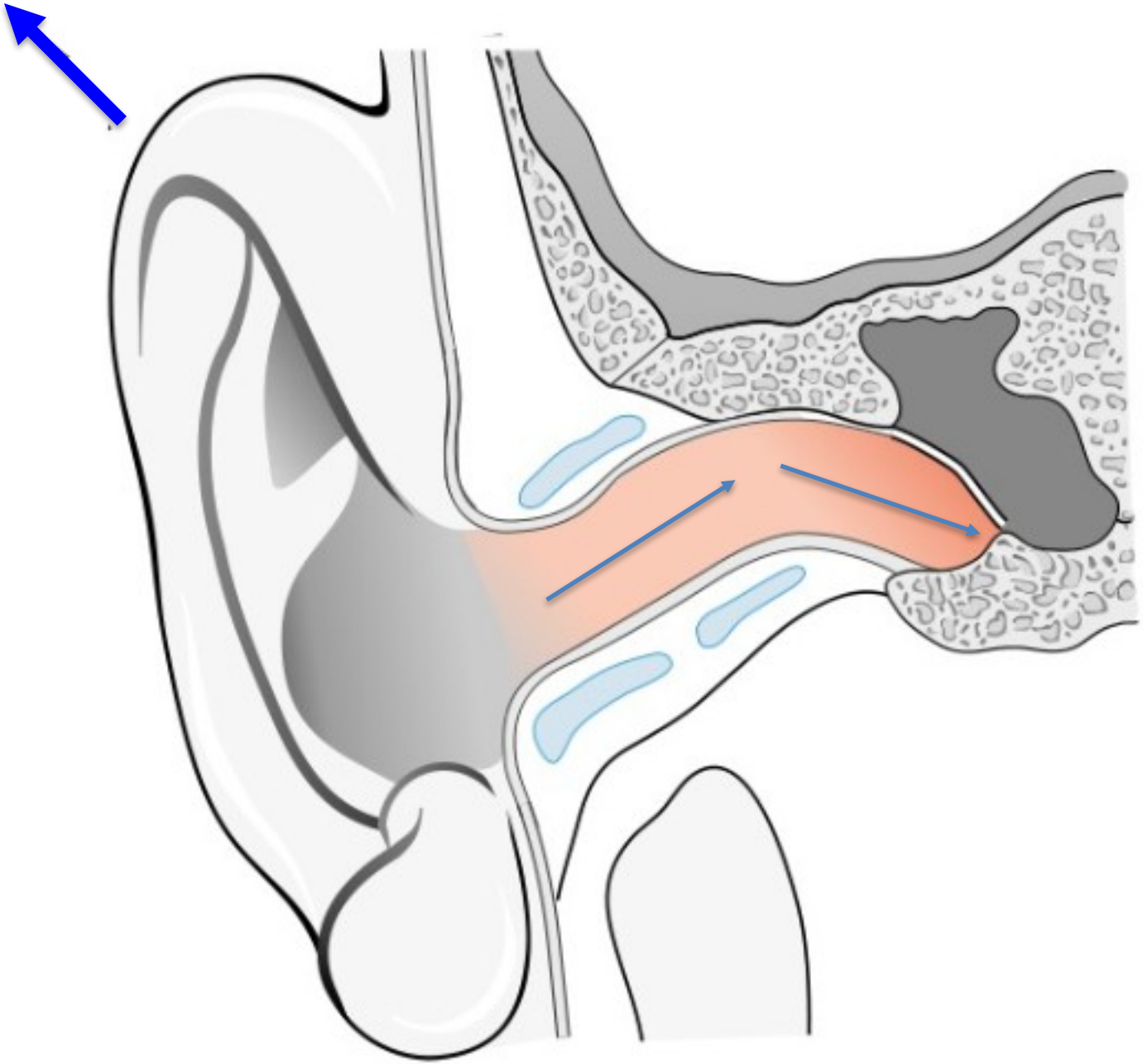
The external acoustic meatus does not have a straight path, and instead travels in an S-shaped curve as follows:

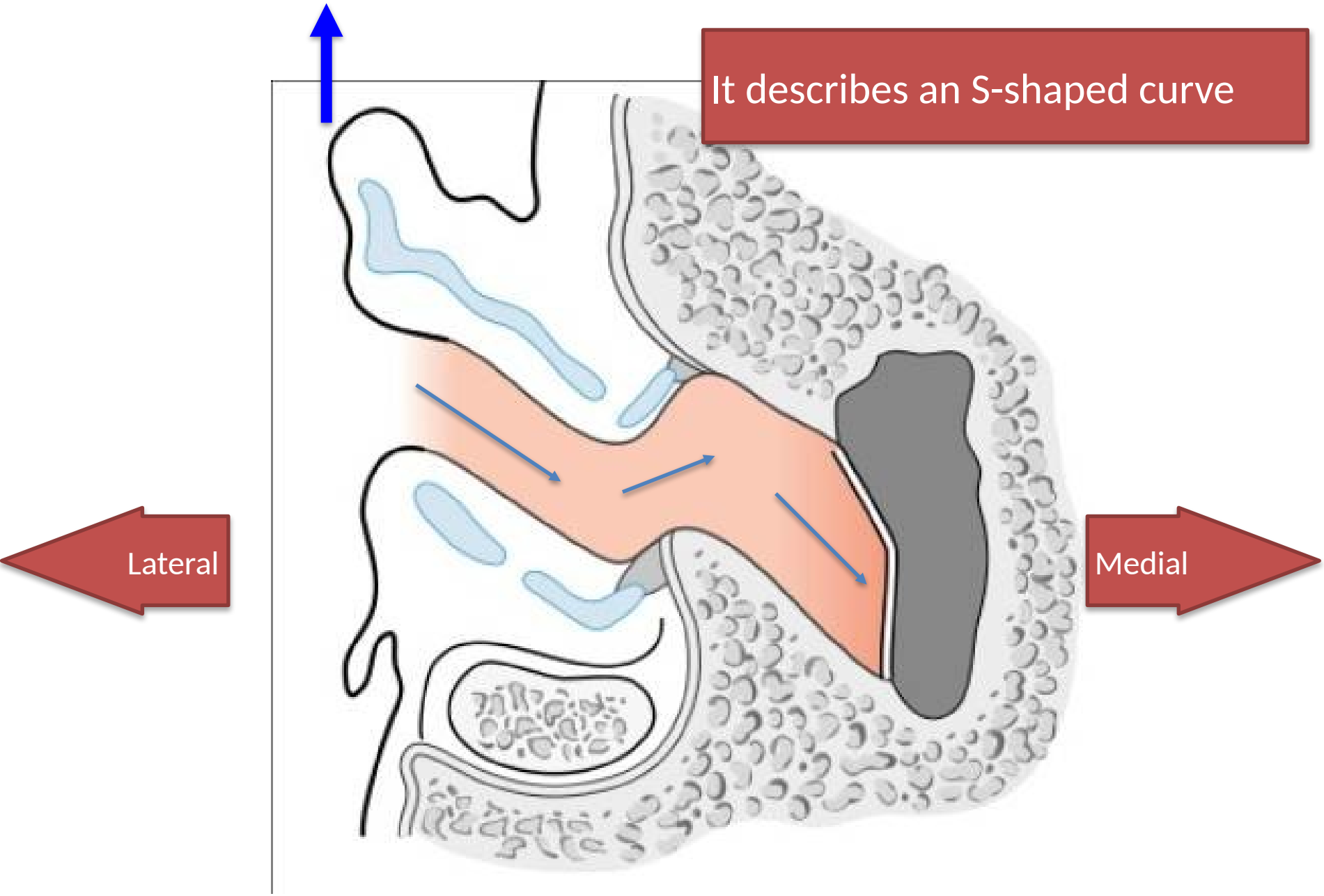
- Initially it travels in a **superoanterior** direction.
- In then turns slightly to move **superoposteriorly**.
- It ends by running in an **inferoanterior** direction.


External ear
Middle ear
Internal ear
Internal acoustic meatus
Auricle
External acoustic meatus
Cartilage
Pharynx
Tympanic membrane
Pharyngotympanic tube



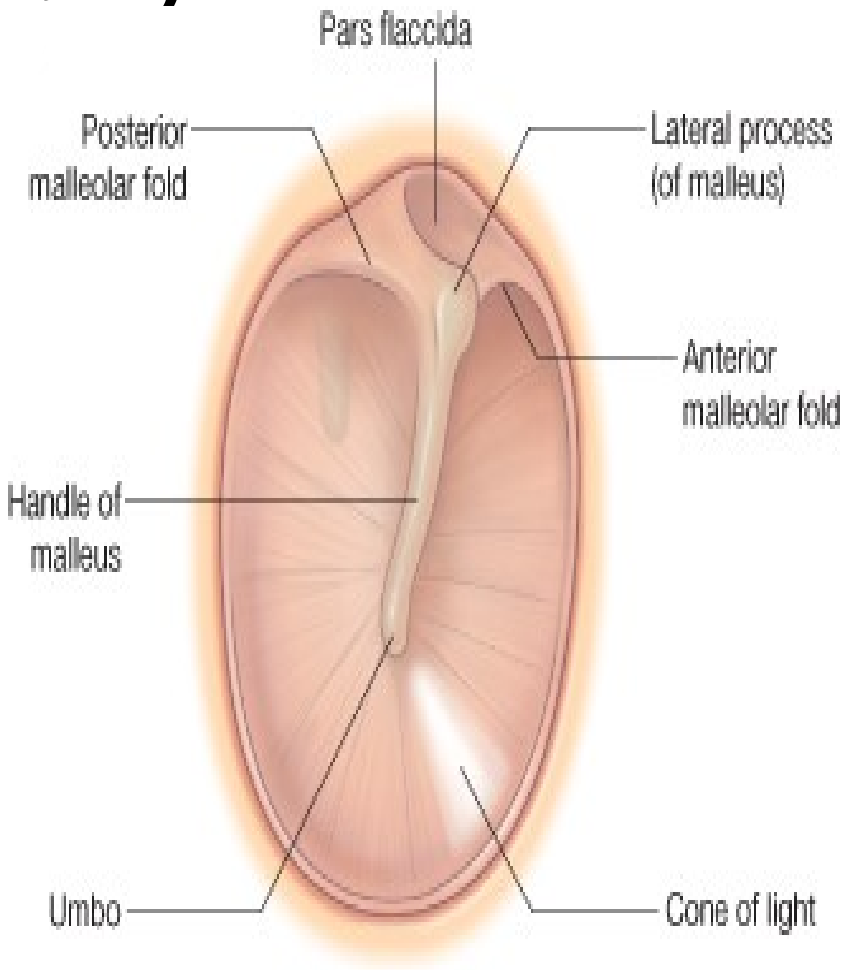
24 mm in length

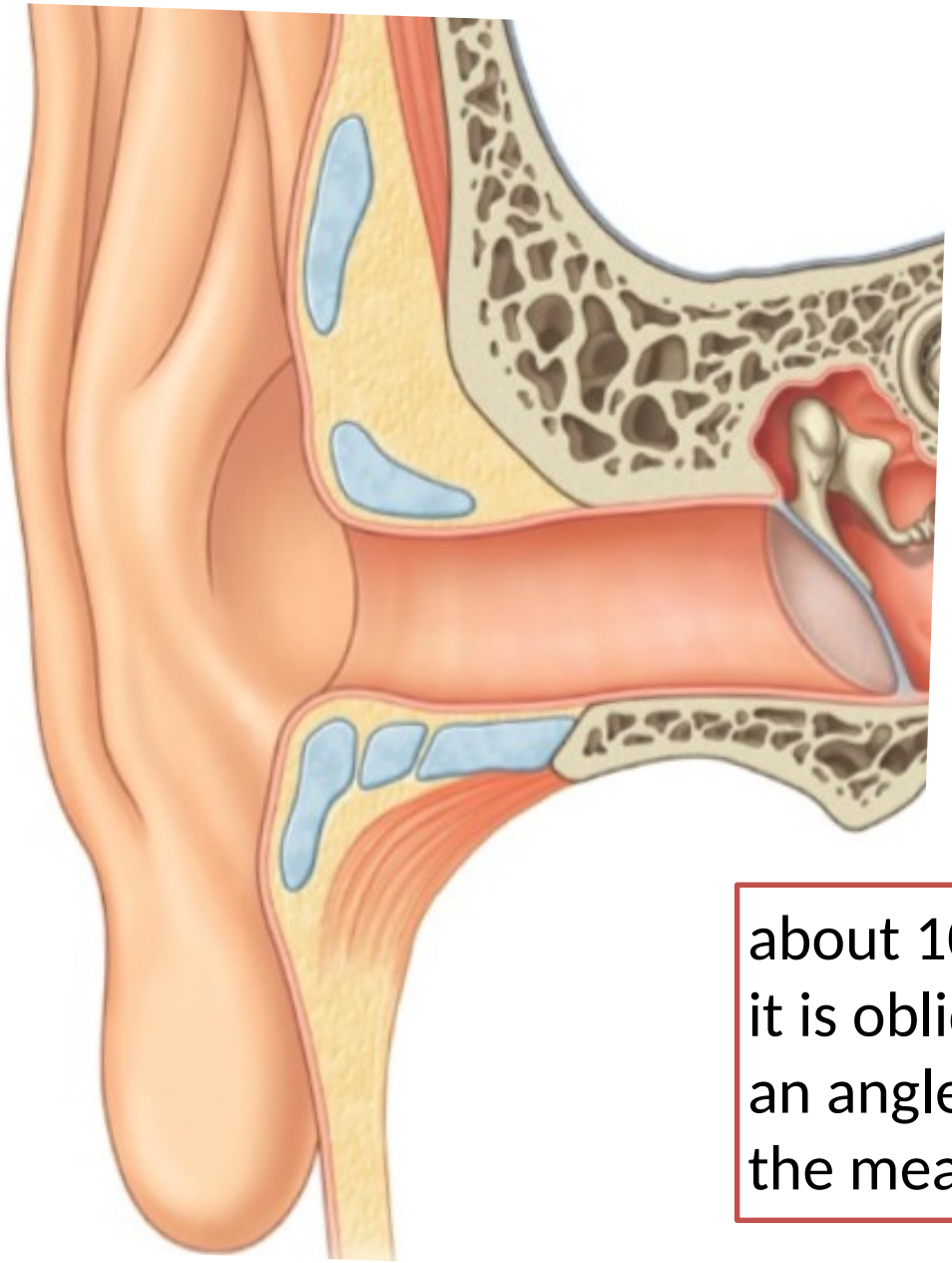



It describes an S-shaped curve
Lateral
Medial

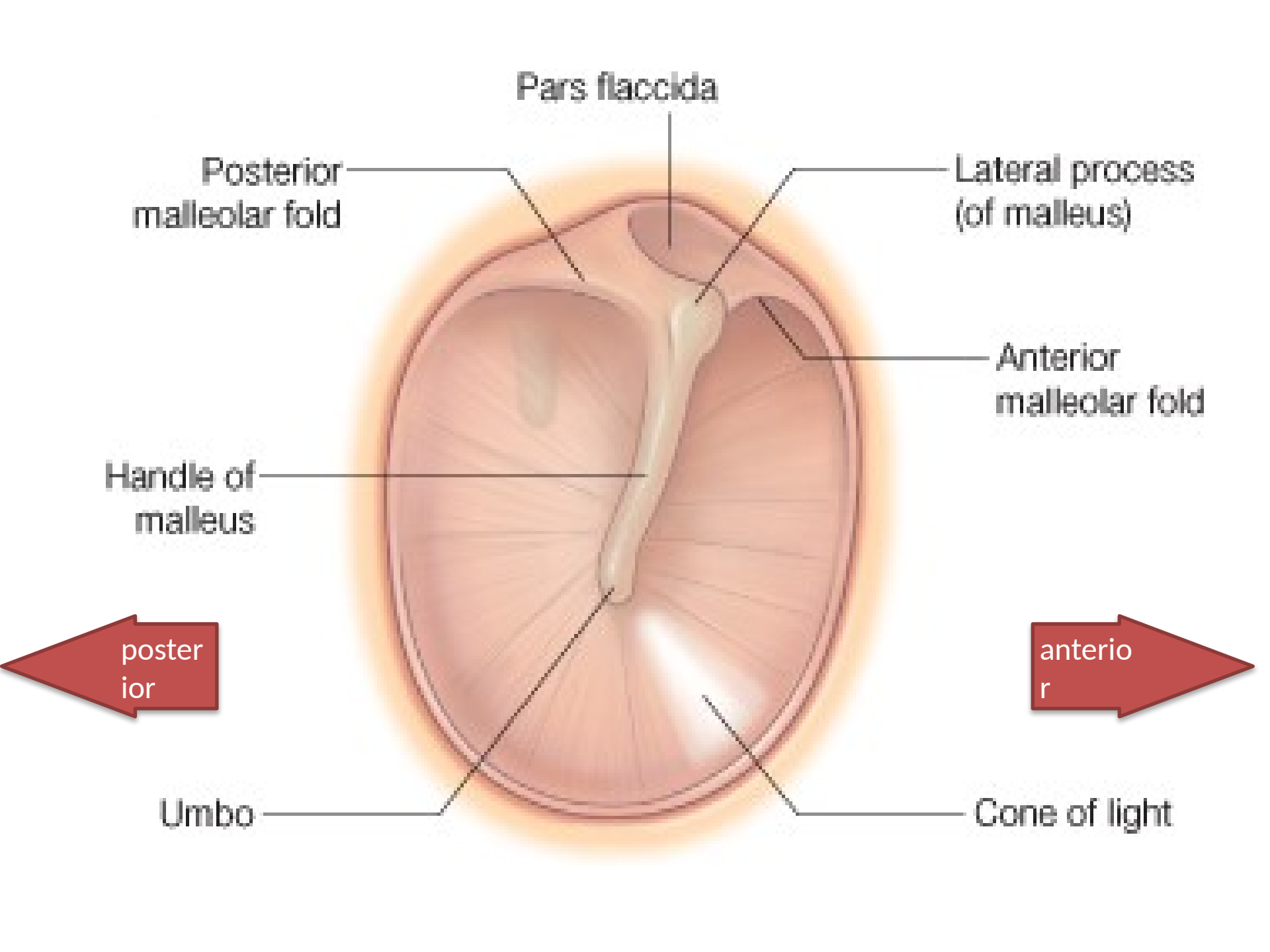
TYMPANIC MEMBRANE (EAR DRᵻJM)

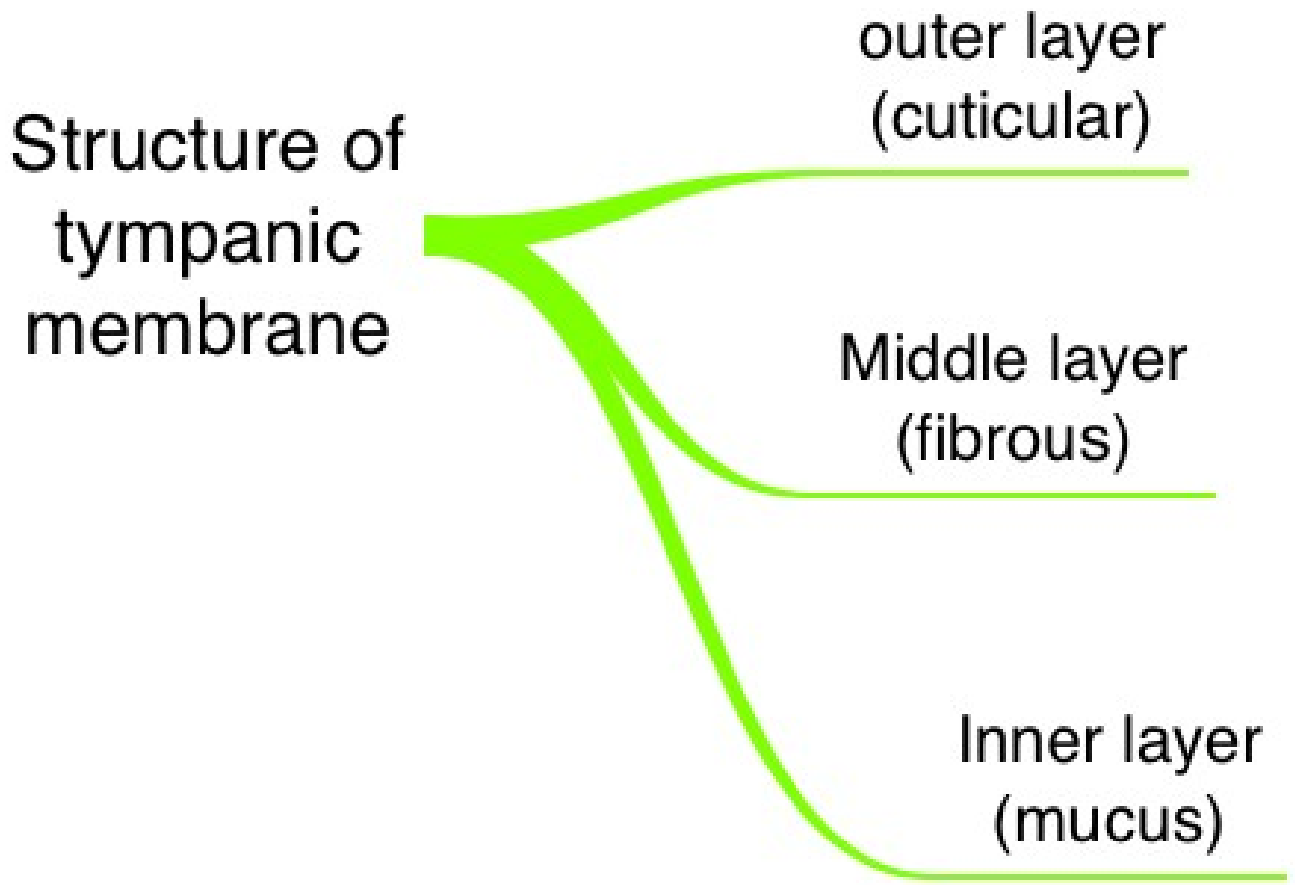
The external acoustic meatus ends at the tympanic membrane. The tympanic membrane has a double layered structure, covered with **skin** on the outside and a **mucous membrane** on the inside. At the core of the membrane is connective tissue. It is connected to the surrounding **temporal** bone by a fibrocartilaginous ring.
The tympanic membrane is **translucent**, therefore structures within the middle ear can be seen. On the inner surface of the membrane, the **handle of malleus** attaches to the tympanic membrane, at a point called the **umbo** of tympanic membrane.
The handle of malleus continues **superiorly**, and at its highest point, a small projection called the **lateral process** of the malleus can be seen. The parts of the tympanic membrane moving away from the lateral process are called the anterior and posterior malleolar **folds.**

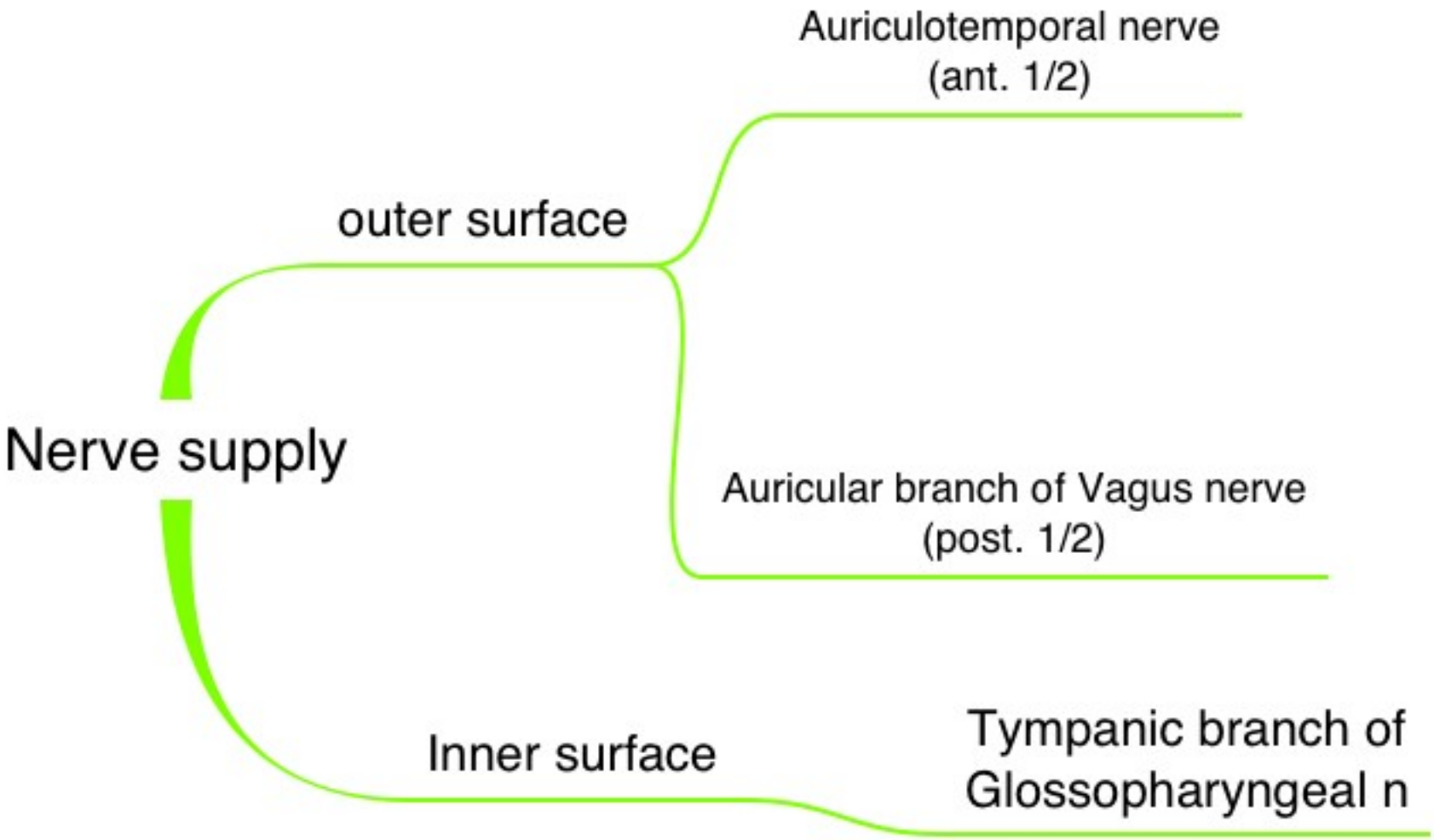


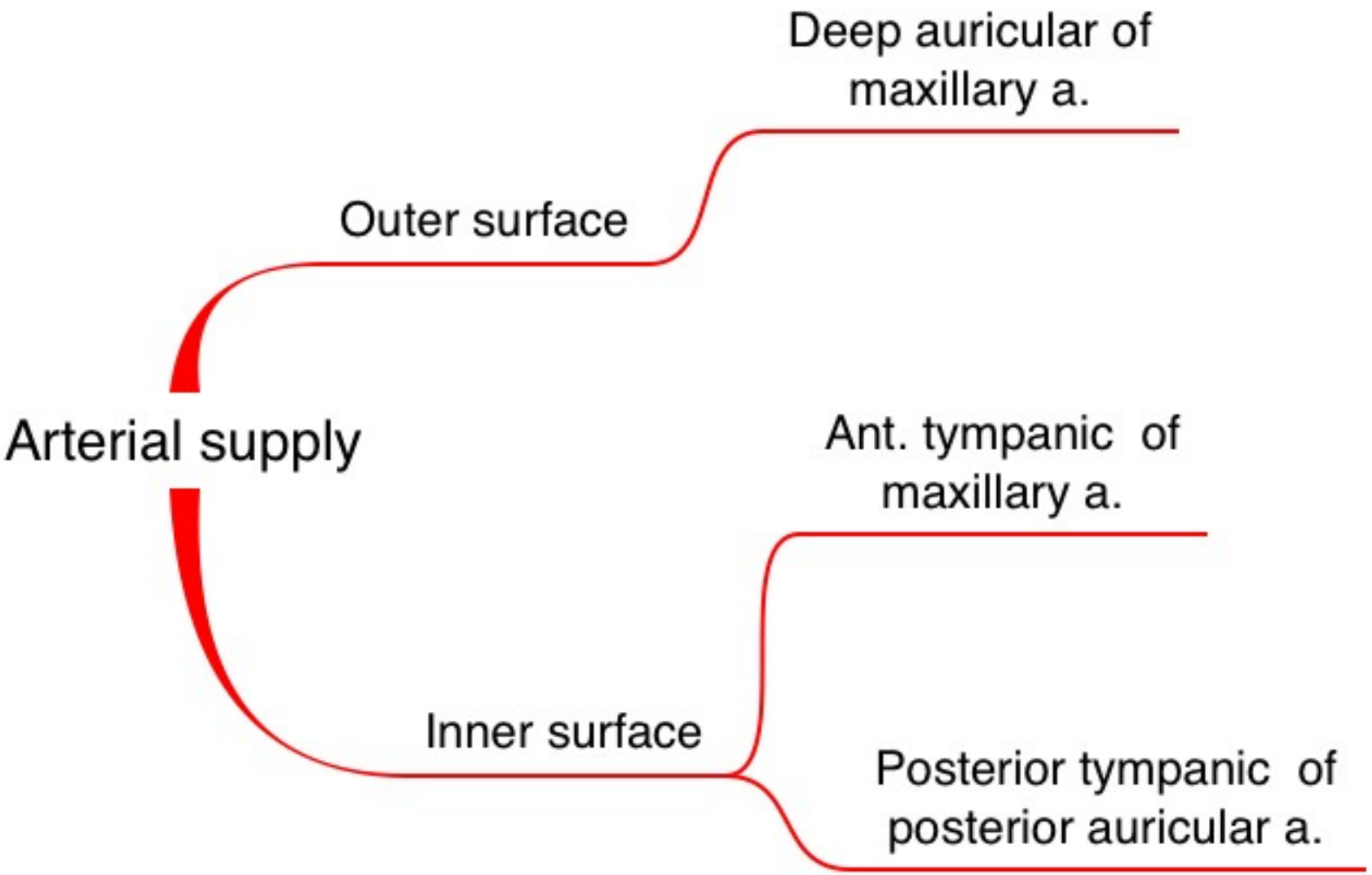


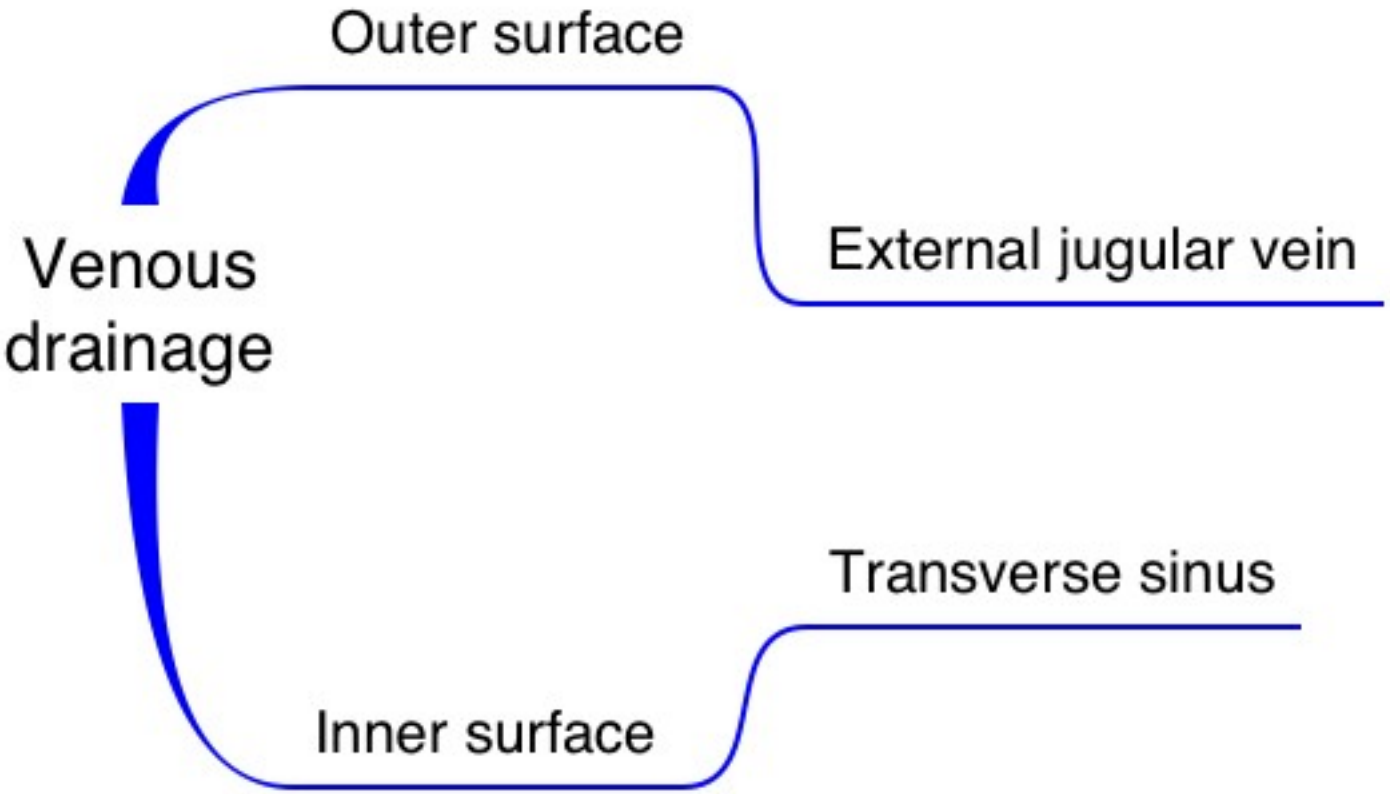
about 10 mm in diameter
it is obliquely directed making an angle of 55 with the floor of the meatus

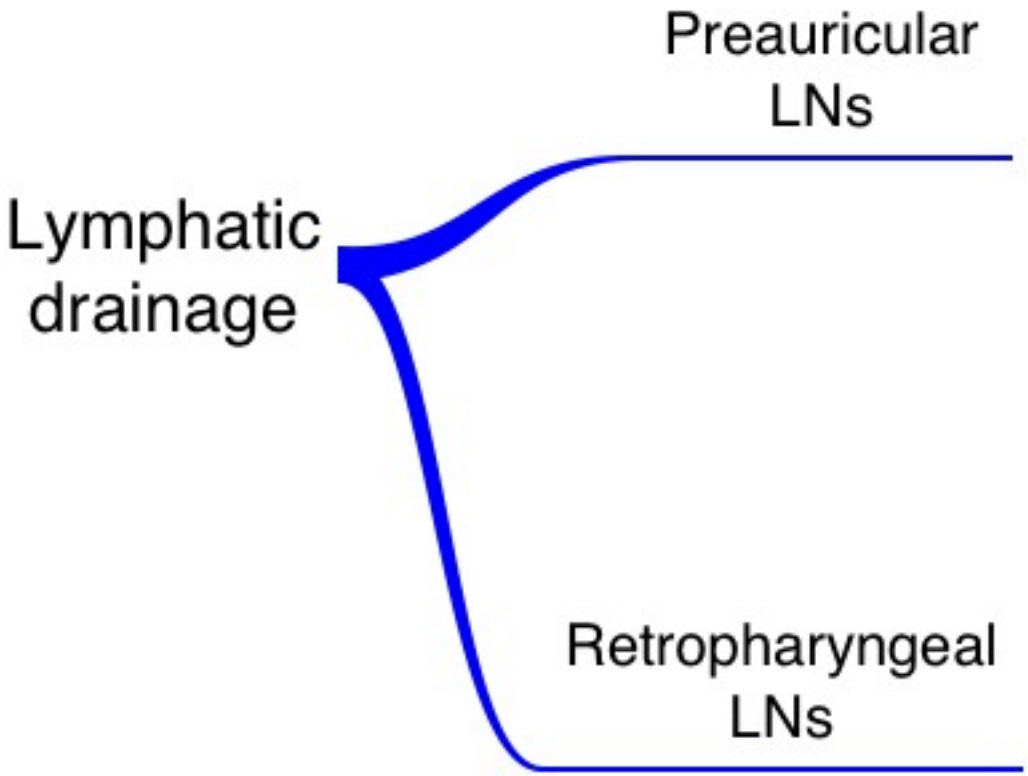

Pars flaccida
Posterior malleolar fold
Lateral process (of malleus)
Anterior malleolar fold
Handle of malleus
posterior
anterior
Umbo
Cone of light


Structure of tympanic membrane
outer layer (cuticular)
Middle layer (fibrous)
Inner layer (mucus)


Nerve supply
outer surface
Auriculotemporal nerve
(ant. 1/2)
Auricular branch of Vagus nerve
(post. 1/2)
Inner surface
Tympanic branch of
Glossopharyngeal n


Arterial supply
Outer surface
Deep auricular of maxillary a.
Inner surface
Ant. tympanic of maxillary a.
Posterior tympanic of posterior auricular a.

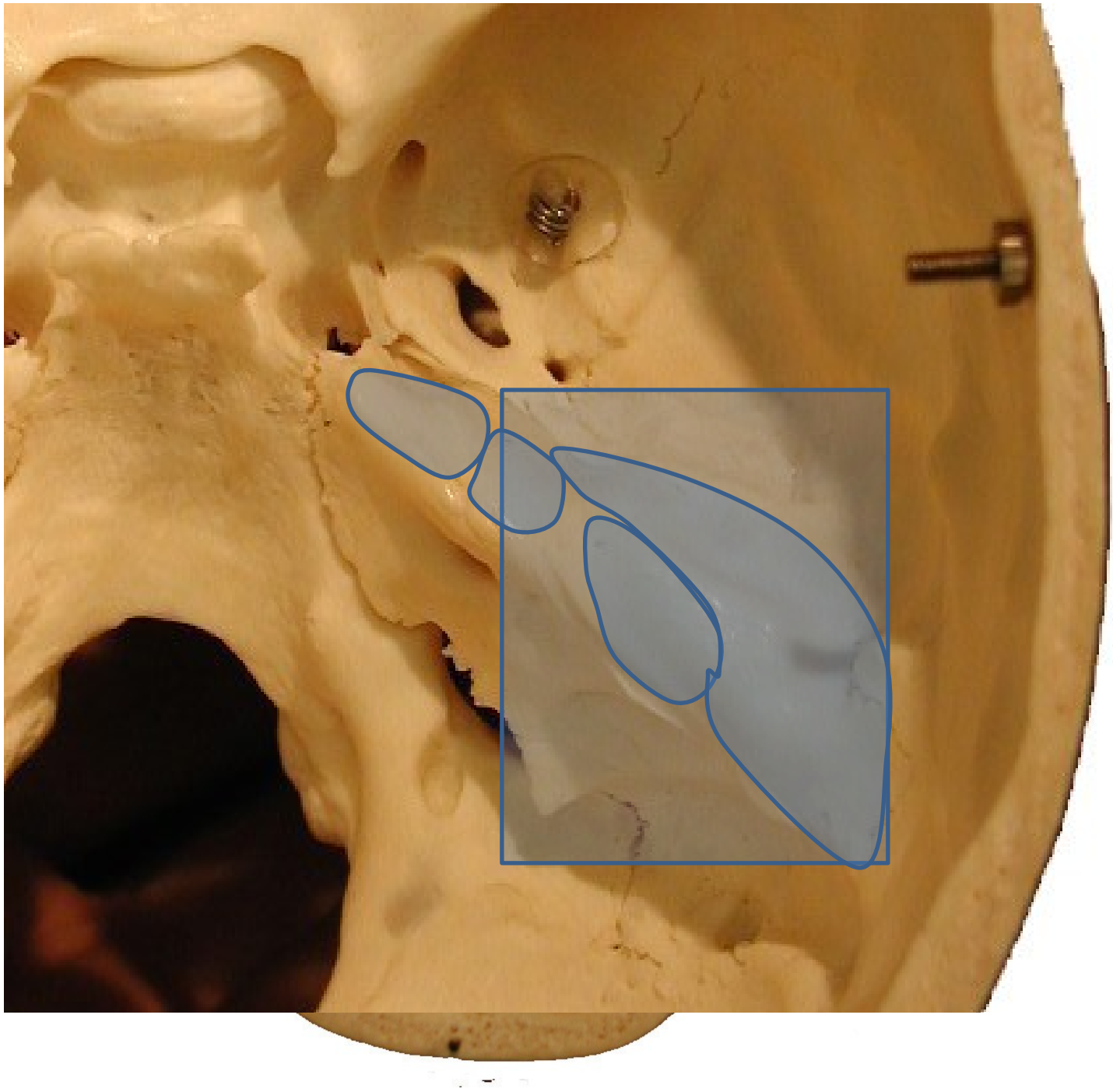

Outer surface
Venous drainage
External jugular vein
Transverse sinus
Inner surface

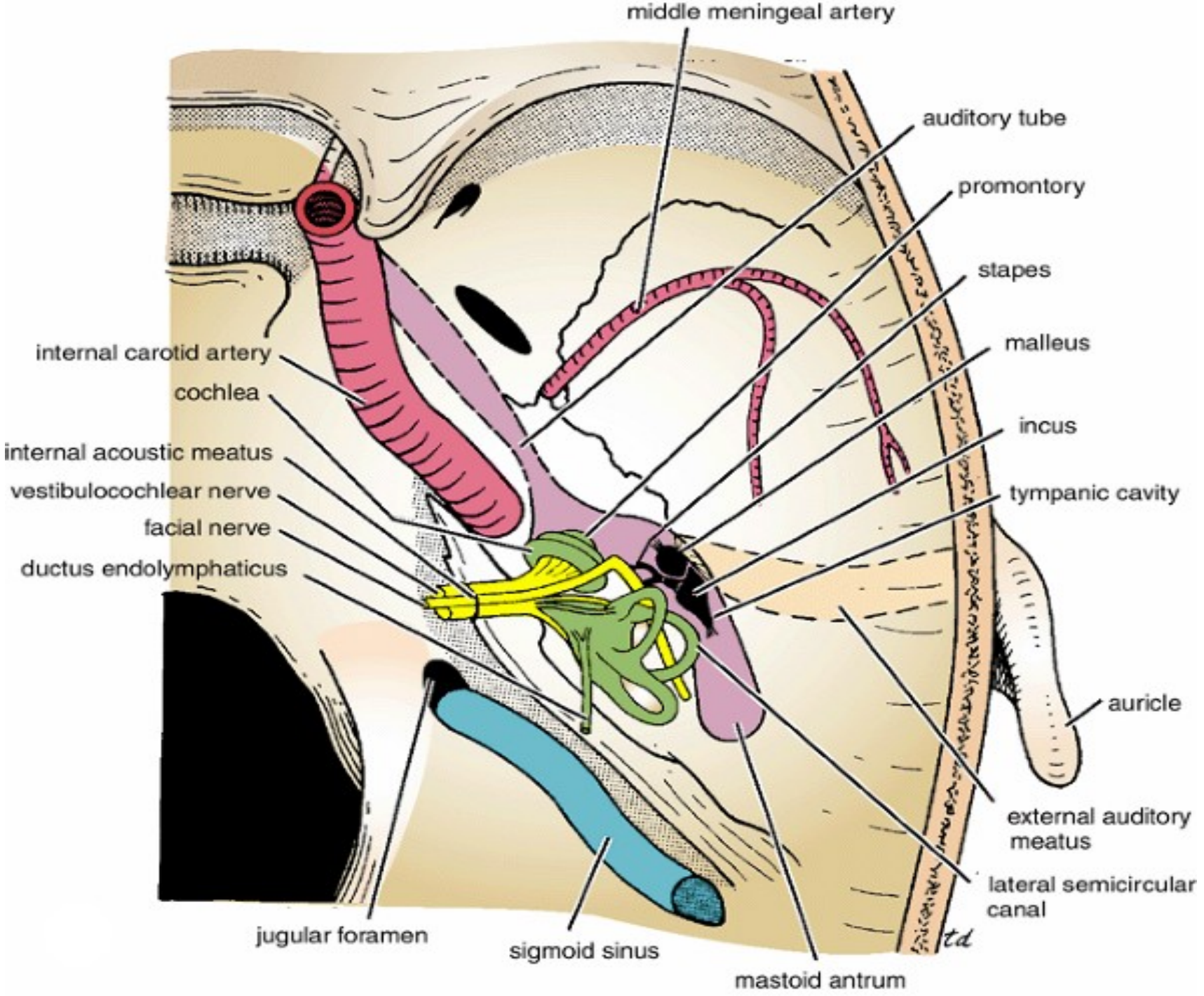

Preauricular
LNs
Lymphatic
drainage
Retropharyngeal
LNs

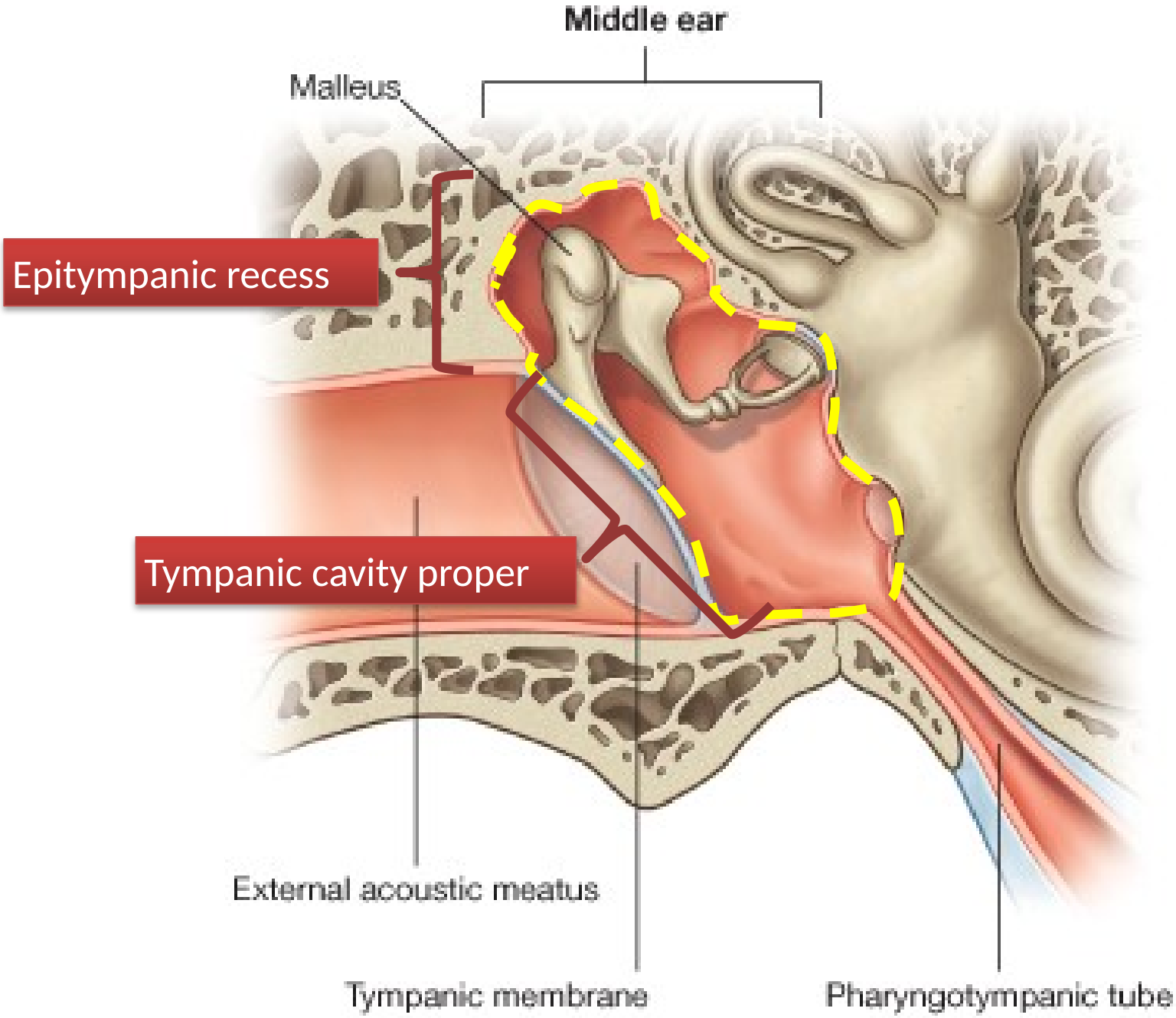
Middle Ear

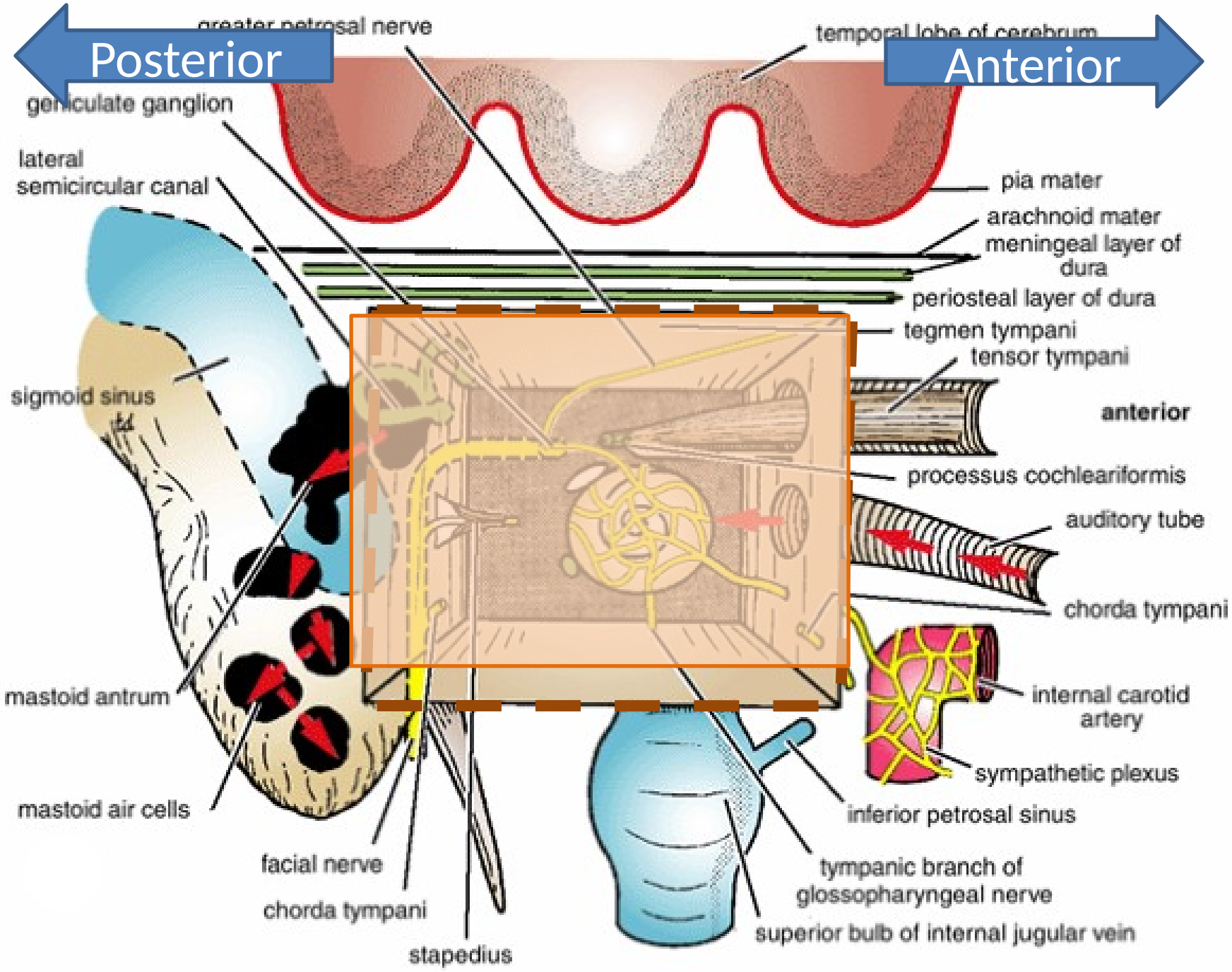
Structure

The middle ear lies within the **temporal** bone, and extends from the **tympanic** membrane to the **lateral** wall of the internal ear. The main function of the middle ear is to **transmit** vibrations from the tympanic membrane to the inner ear – it does this via the three bones of the ear.



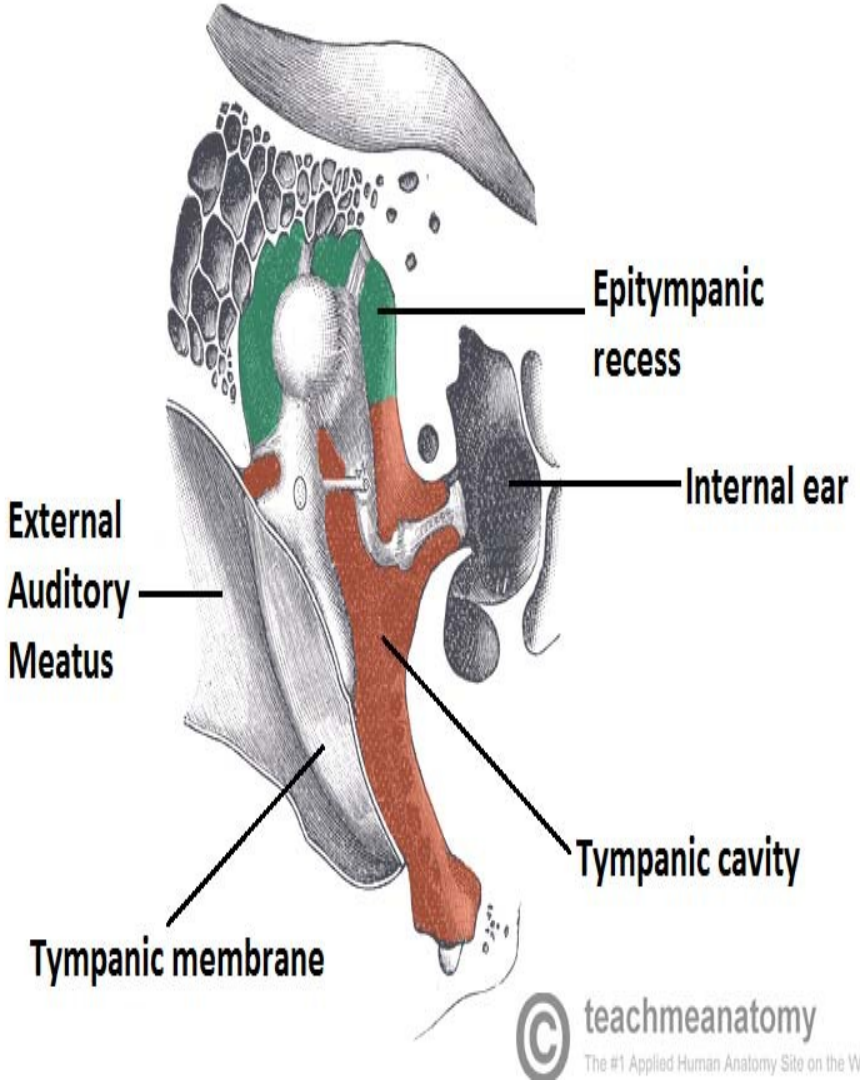

middle meningeal artery
auditory tube
promontory
stapes
malleus
incus
tympanic cavity
internal carotid artery
cochlea
internal acoustic meatus
vestibulocochlear nerve
facial nerve
ductus endolymphaticus
auricle
external auditory meatus
lateral semicircular canal
jugular foramen
sigmoid sinus
mastoid antrum


Middle ear
Malleus
Epitympanic recess
Tympanic cavity proper
External acoustic meatus
Tympanic membrane
Pharyngotympanic tube


Posterior
Anterior
greater petrosal nerve
temporal lobe of cerebrum
geniculate ganglion
lateral semicircular canal
pia mater
arachnoid mater
meningeal layer of dura
periosteal layer of dura
tegmen tympani
tensor tympani
sigmoid sinus
anterior
processus cochleariformis
auditory tube
chorda tympani
mastoid antrum
internal carotid artery
sympathetic plexus
mastoid air cells
inferior petrosal sinus
facial nerve
tympanic branch of glossopharyngeal nerve
chorda tympani
superior bulb of internal jugular vein
stapedius

PARTS OF MIDDLE EAR

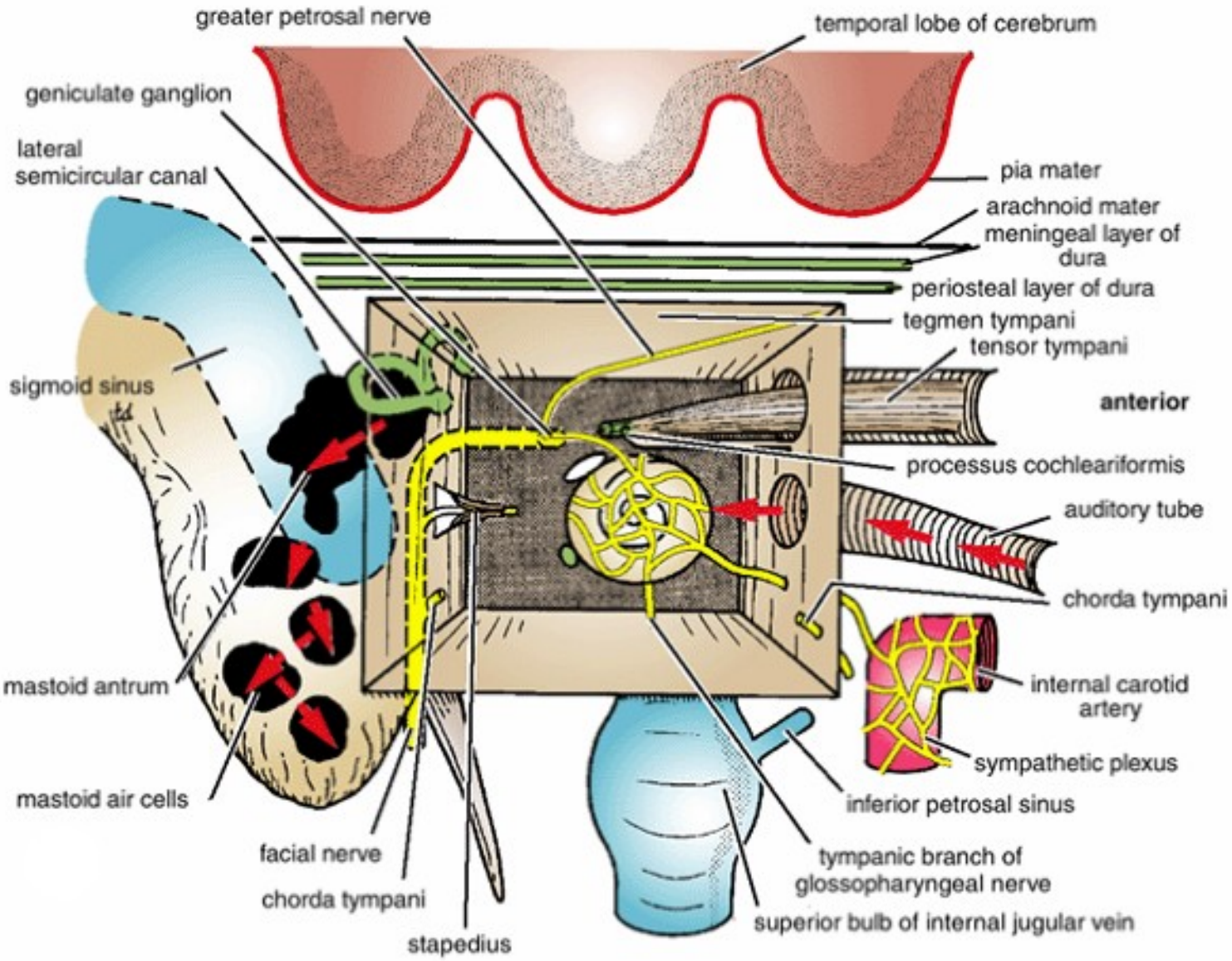
The middle ear can be split into two; the **tympanic cavity** and **epitympanic recess**. The tympanic cavity lies medially to the tympanic membrane. It contains the majority of the **bones** of the middle ear. The epitympanic recess is found superiorly, near the mastoid **air cells**.



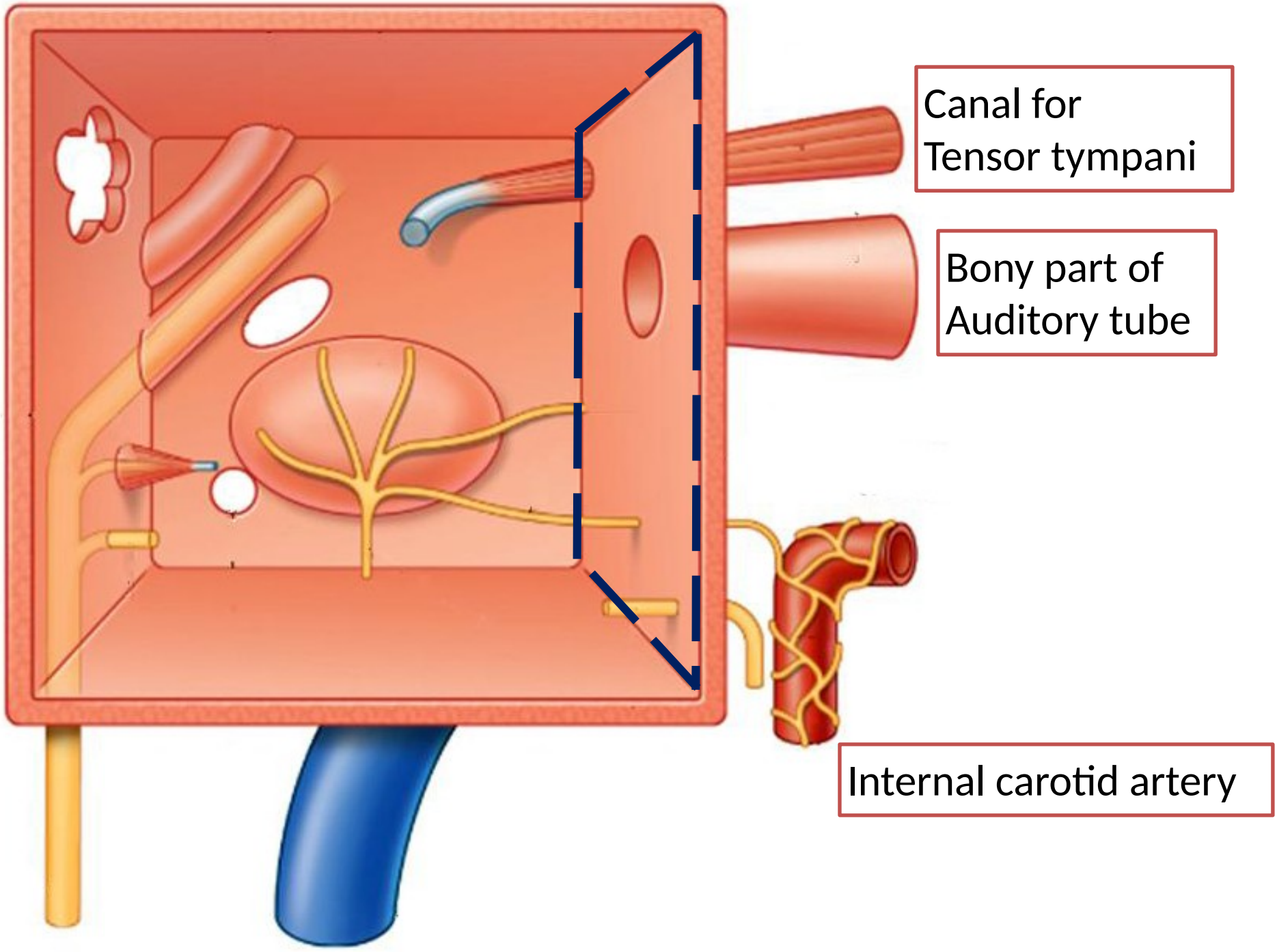
BORDERS

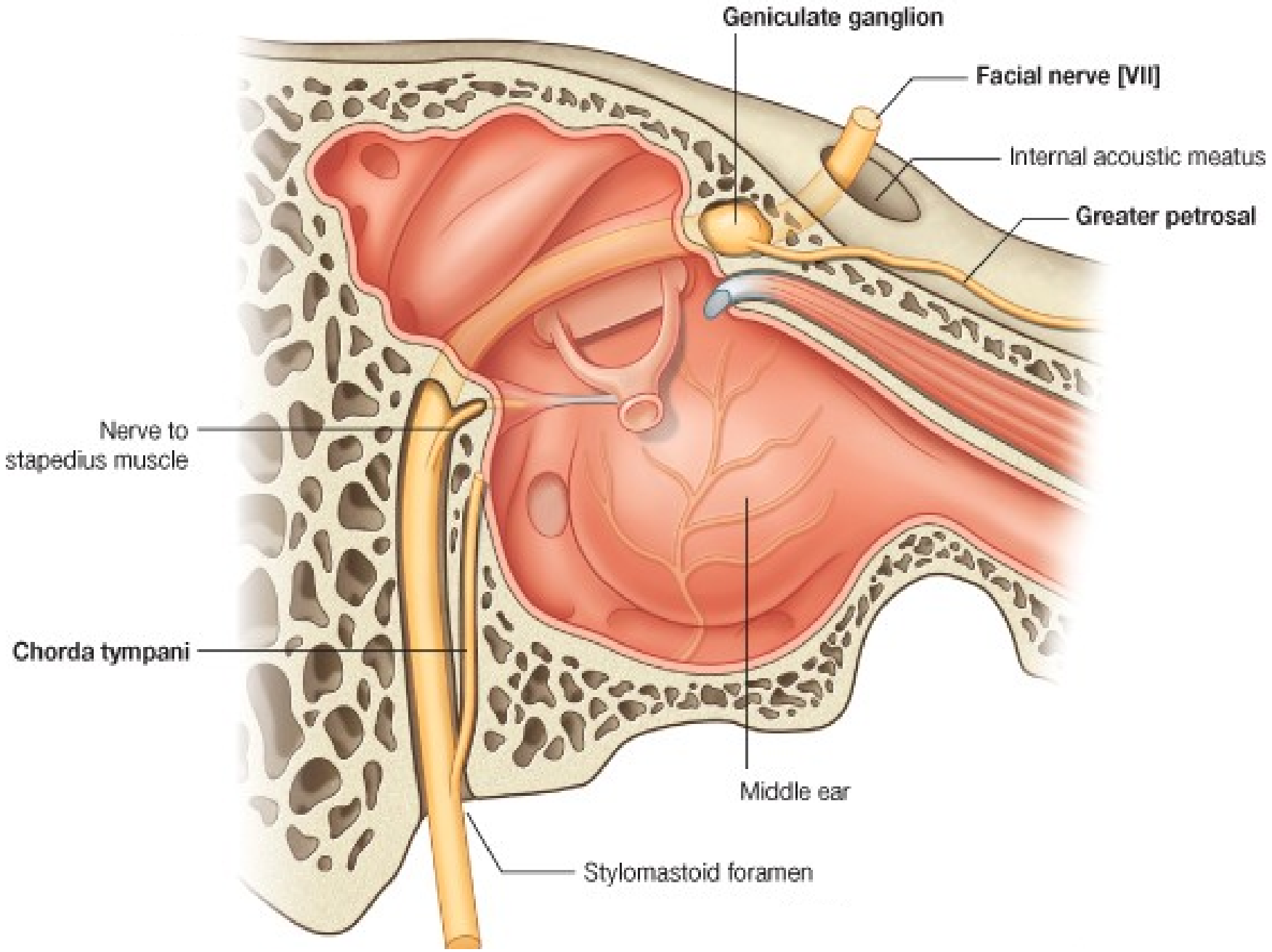
The middle ear can be visualised as a rectangular box, with a roof and floor, medial and lateral walls and anterior and posterior walls.

- **Roof** – Formed by a thin bone from the **petrous** part of the temporal bone. It separates the middle ear from the middle cranial fossa.
- **Floor** – Known as the jugular wall, it consists of a thin layer of bone, which separates the middle ear from the internal jugular vein
- **Lateral Wall** – This is made up of the tympanic membrane and the lateral wall of the epitympanic recess.
- **Medial Wall** – Formed by the lateral wall of the internal ear. It contains a prominent bulge, produced by the facial nerve as it travels nearby.
- **Anterior Wall** – The anterior wall is a thin bony plate with two openings; for the auditory tube and the tensor tympani muscle. It separates the middle ear from the internal carotid artery.
- **Posterior Wall** – Also known as the mastoid wall, it consists of a bony partition between the tympanic cavity and the mastoid air cells. Superiorly, there is a hole in this partition, allowing the two areas to communication. This hole is known as the **aditus** to the mastoid antrum.


greater petrosal nerve
temporal lobe of cerebrum
geniculate ganglion
lateral
semicircular canal
pia mater
arachnoid mater
meningeal layer of
dura
periosteal layer of dura
tegmen tympani
tensor tympani
sigmoid sinus
anterior
processus cochleariformis
auditory tube
chorda tympani
internal carotid
artery
mastoid antrum
sympathetic plexus
mastoid air cells
inferior petrosal sinus
facial nerve
tympanic branch of
glossopharyngeal nerve
chorda tympani
superior bulb of internal jugular vein
stapedius

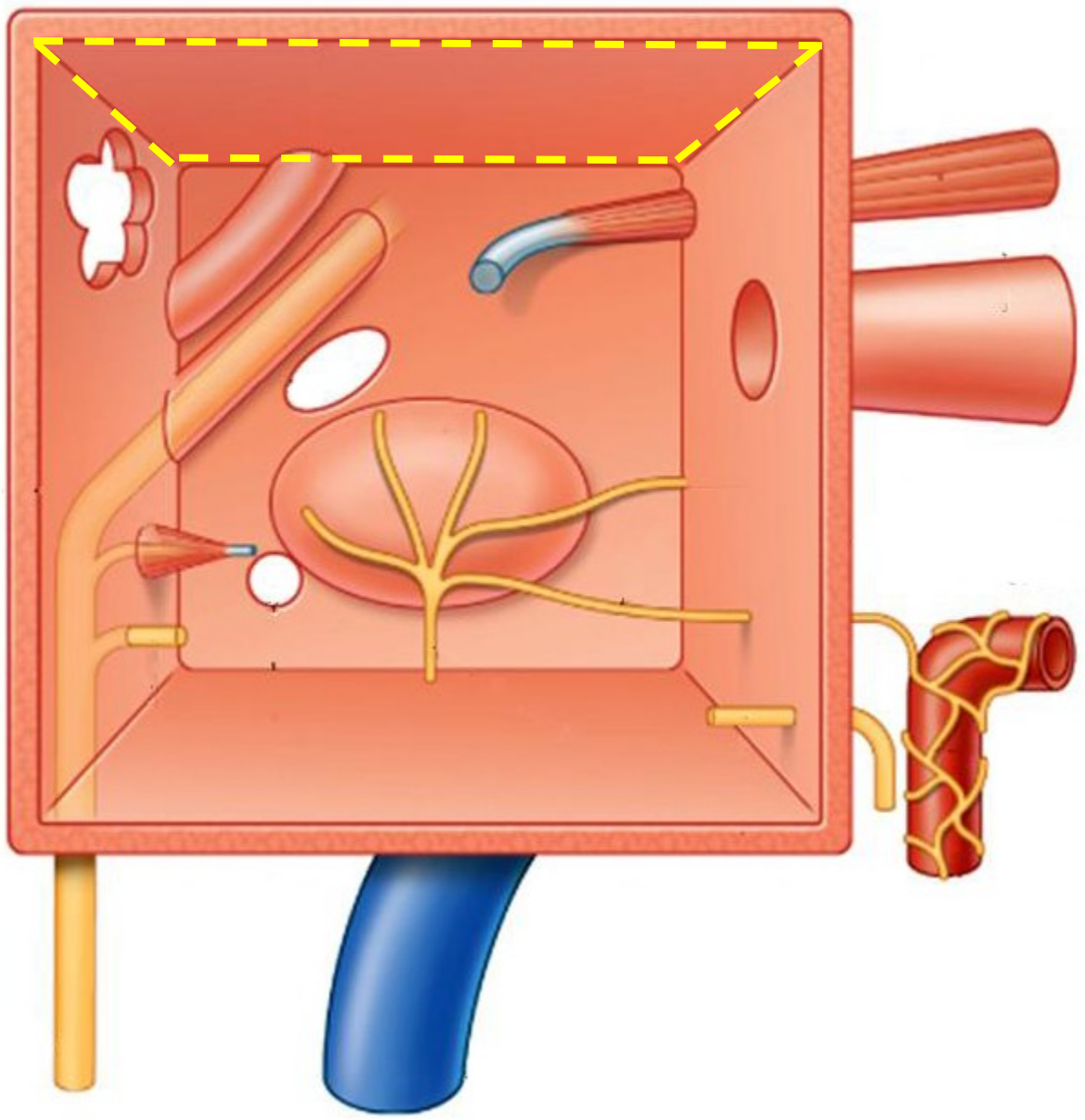
Anterior (carotid) wall


Canal for
Tensor tympani
Bony part of
Auditory tube
Internal carotid artery

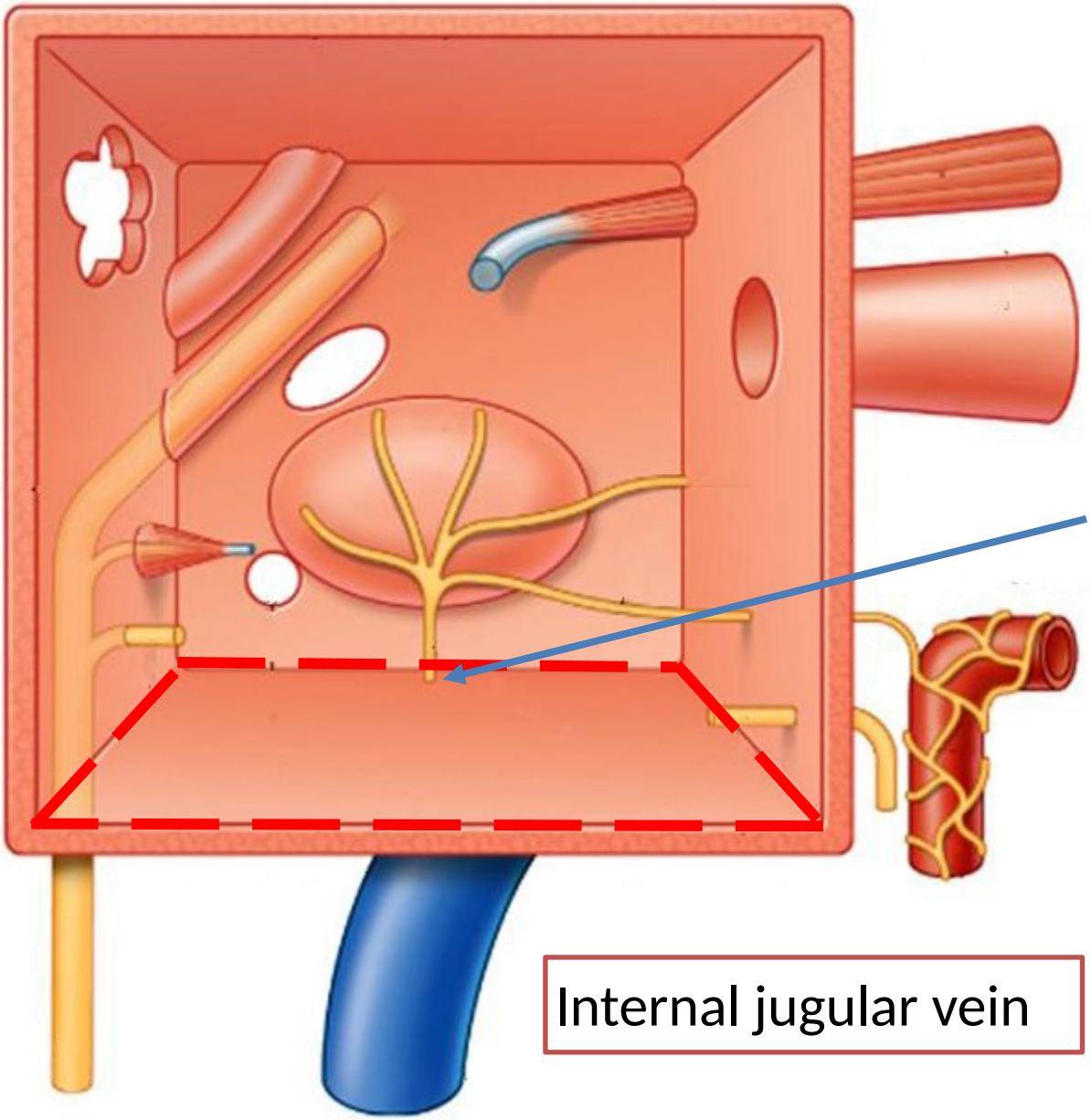

Geniculate ganglion
Facial nerve [VII]
Internal acoustic meatus
Greater petrosal
Nerve to stapedius muscle
Chorda tympani
Middle ear
Stylomastoid foramen

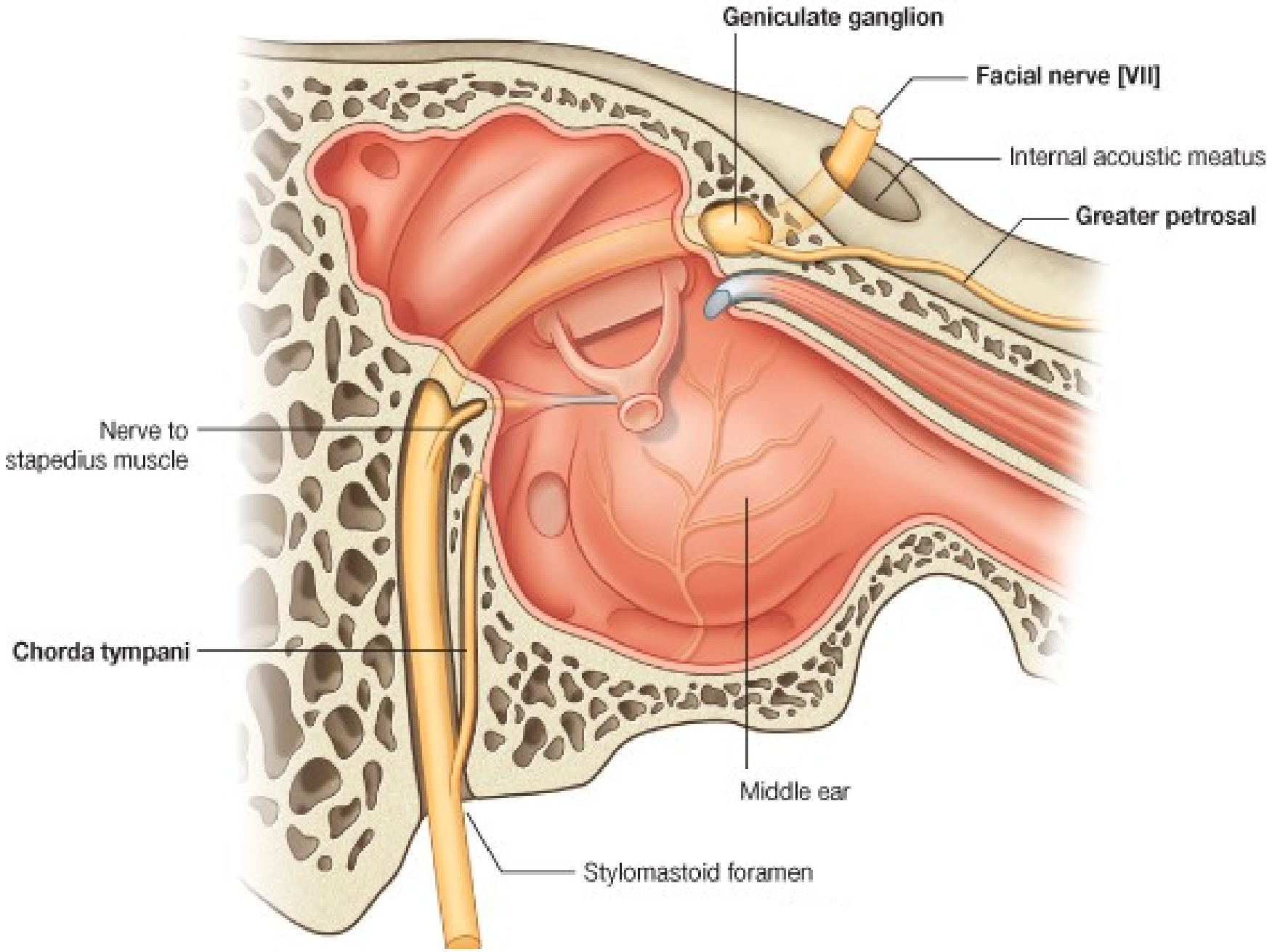
Posterior(mastoid) wall

Tegmen tympani

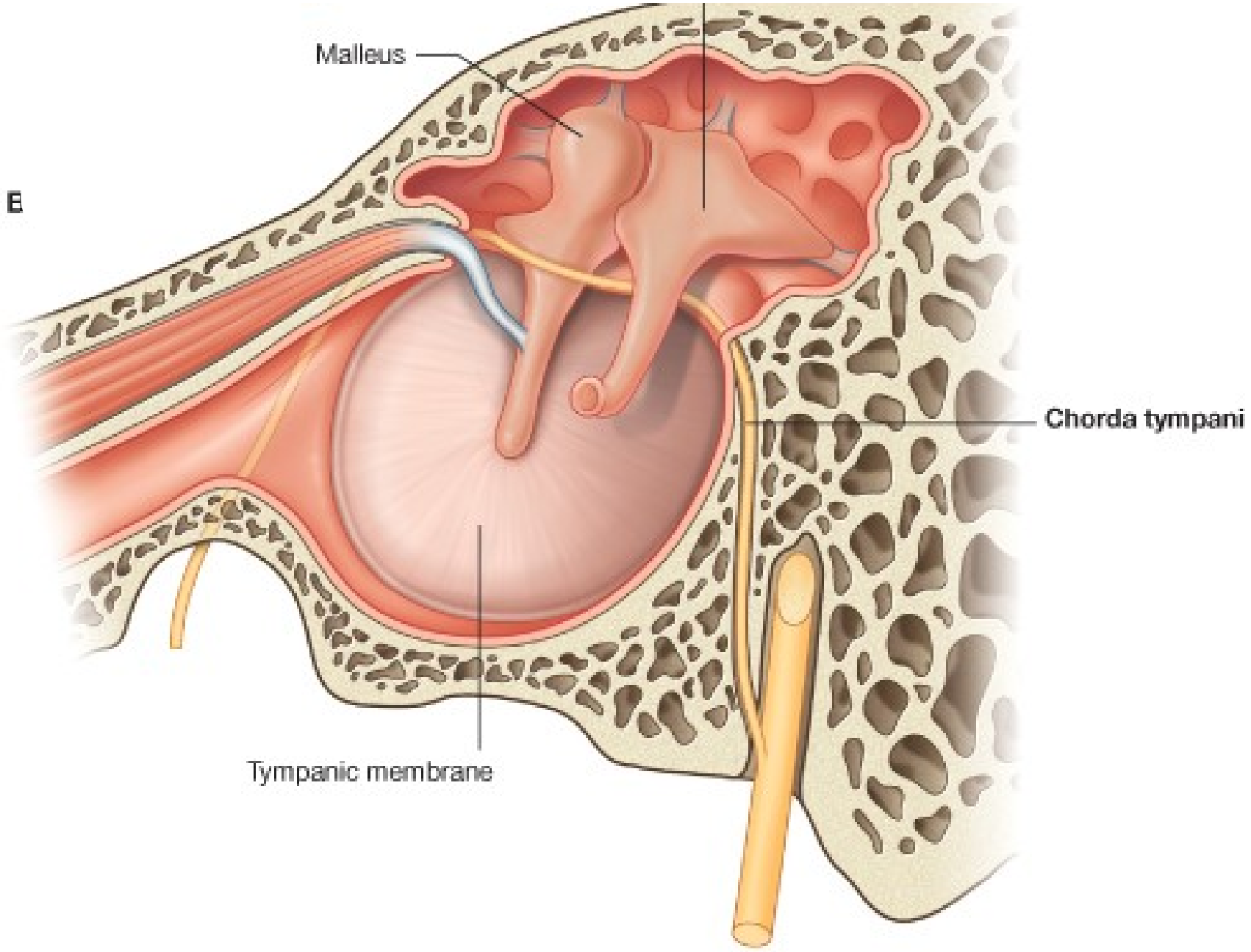


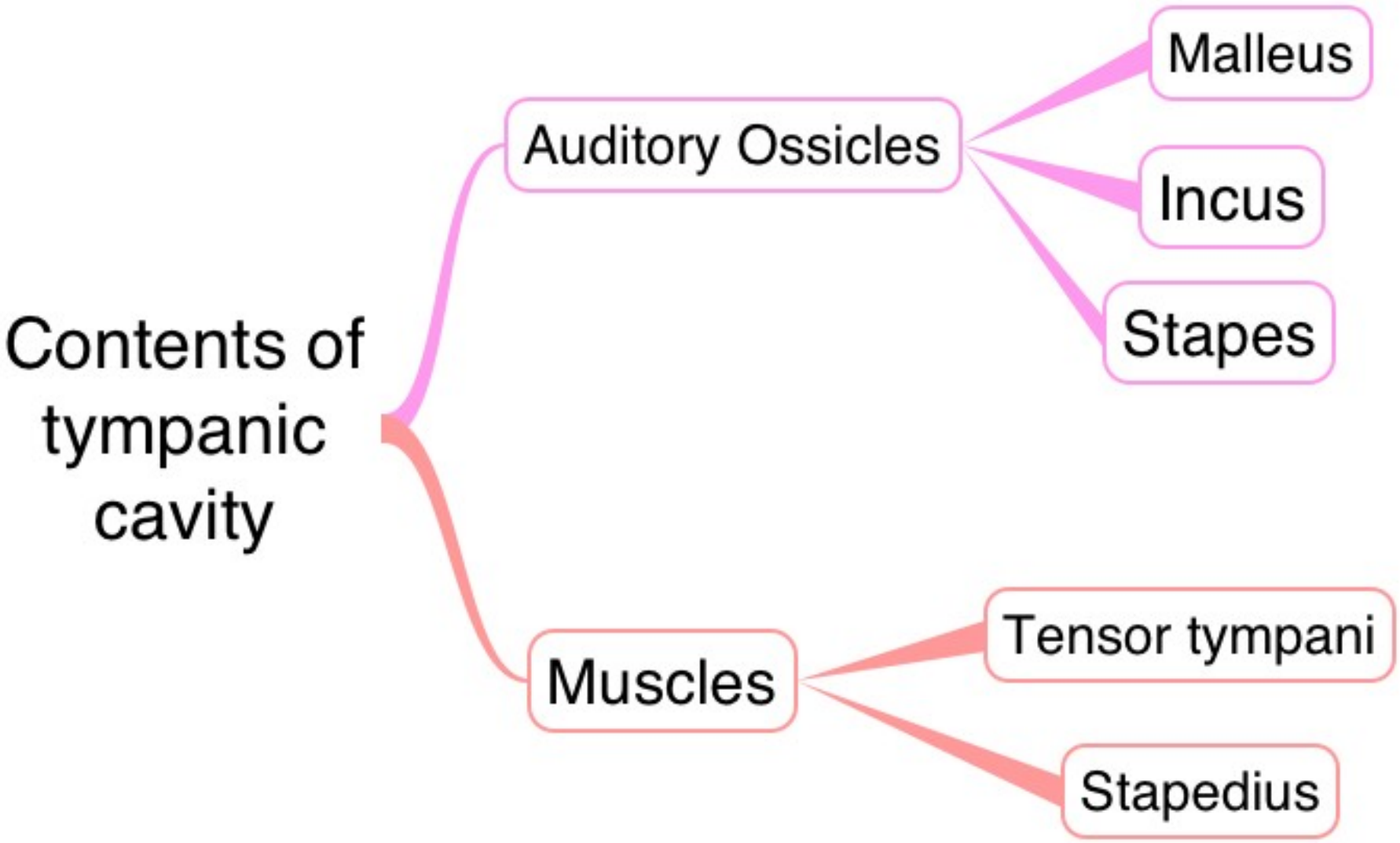
Floor (Jugular) wall


Tympanic branch of XI
Internal jugular vein


Geniculate ganglion
Facial nerve [VII]
Internal acoustic meatus
Greater petrosal
Nerve to stapedius muscle
Chorda tympani
Middle ear
Stylomastoid foramen

Lateral (membranous) wall

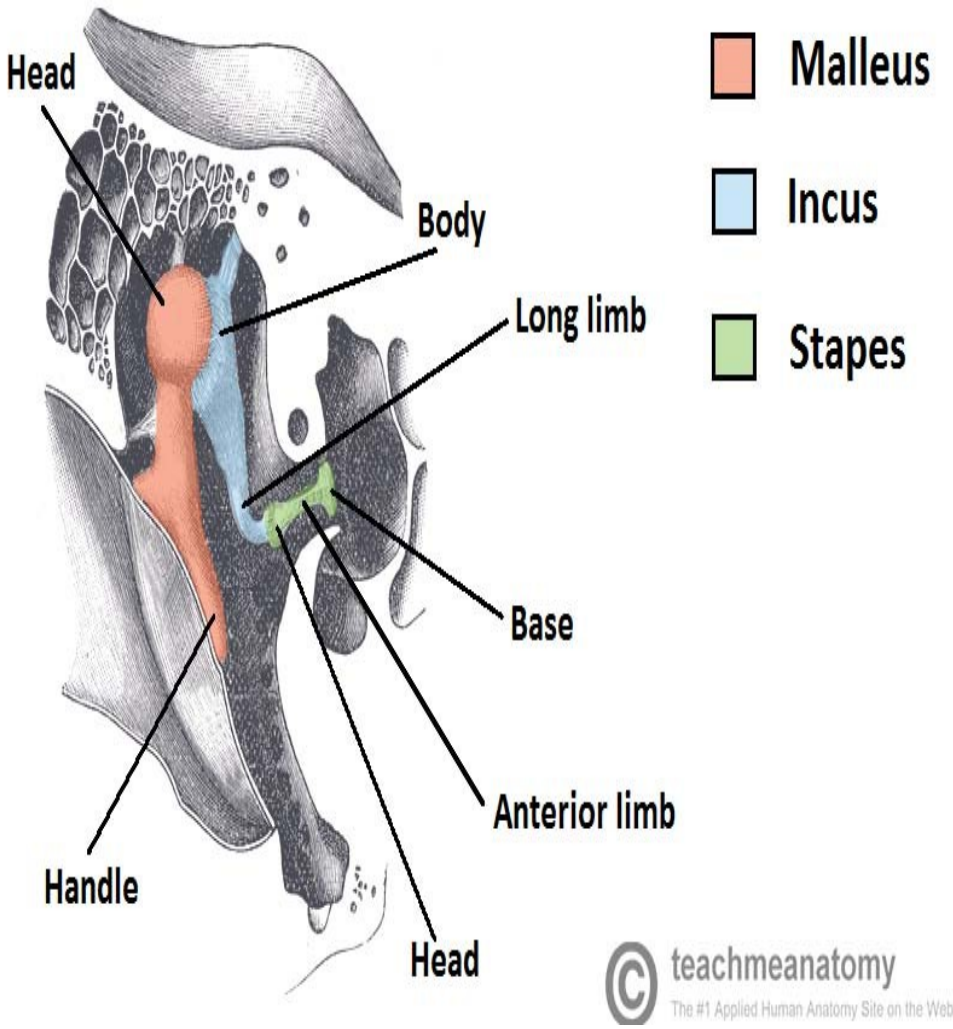

E
Malleus
Chorda tympani
Tympanic membrane

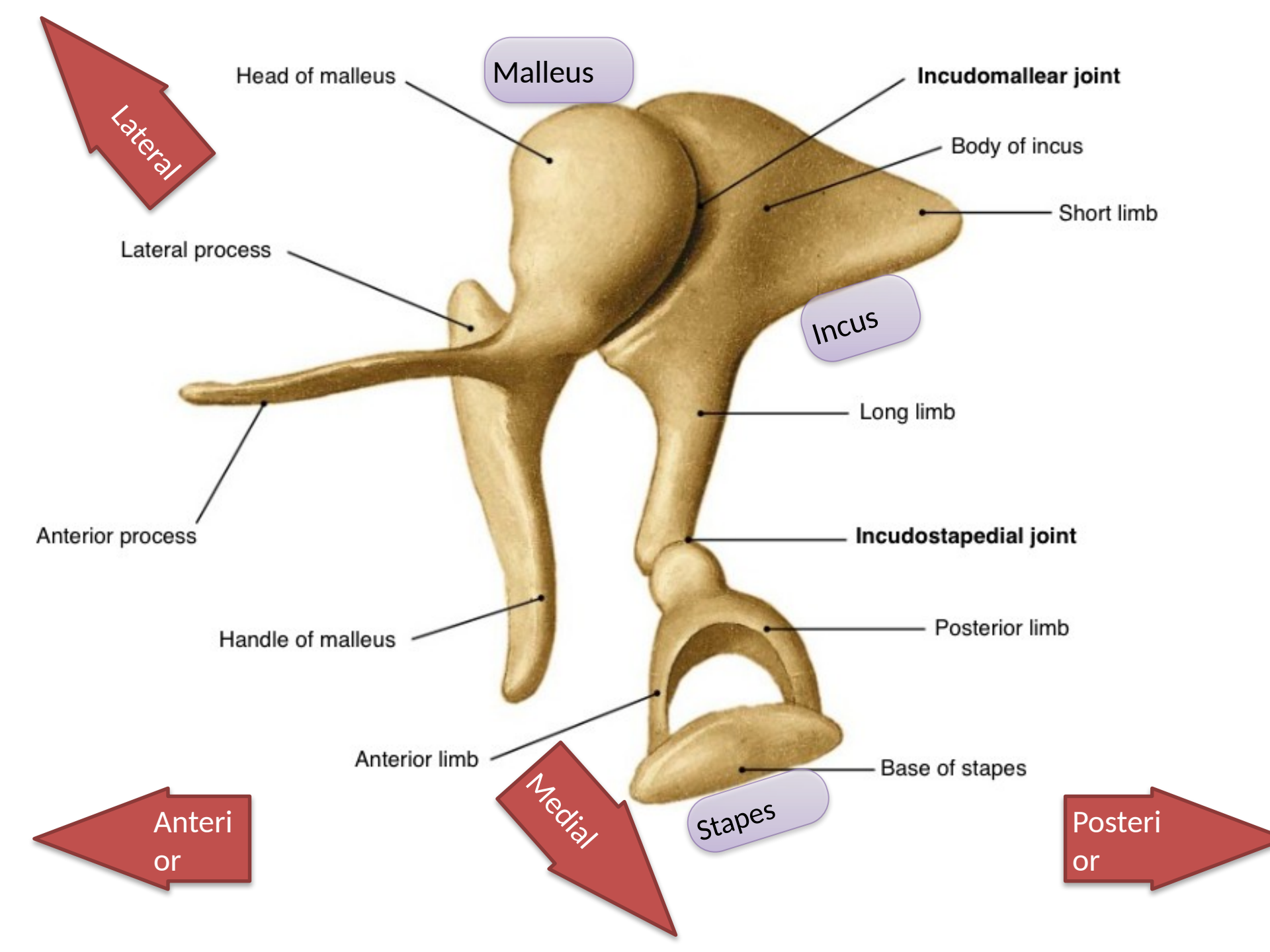

Contents of tympanic cavity
Auditory Ossicles
Malleus
Incus
Stapes
Muscles
Tensor tympani
Stapedius

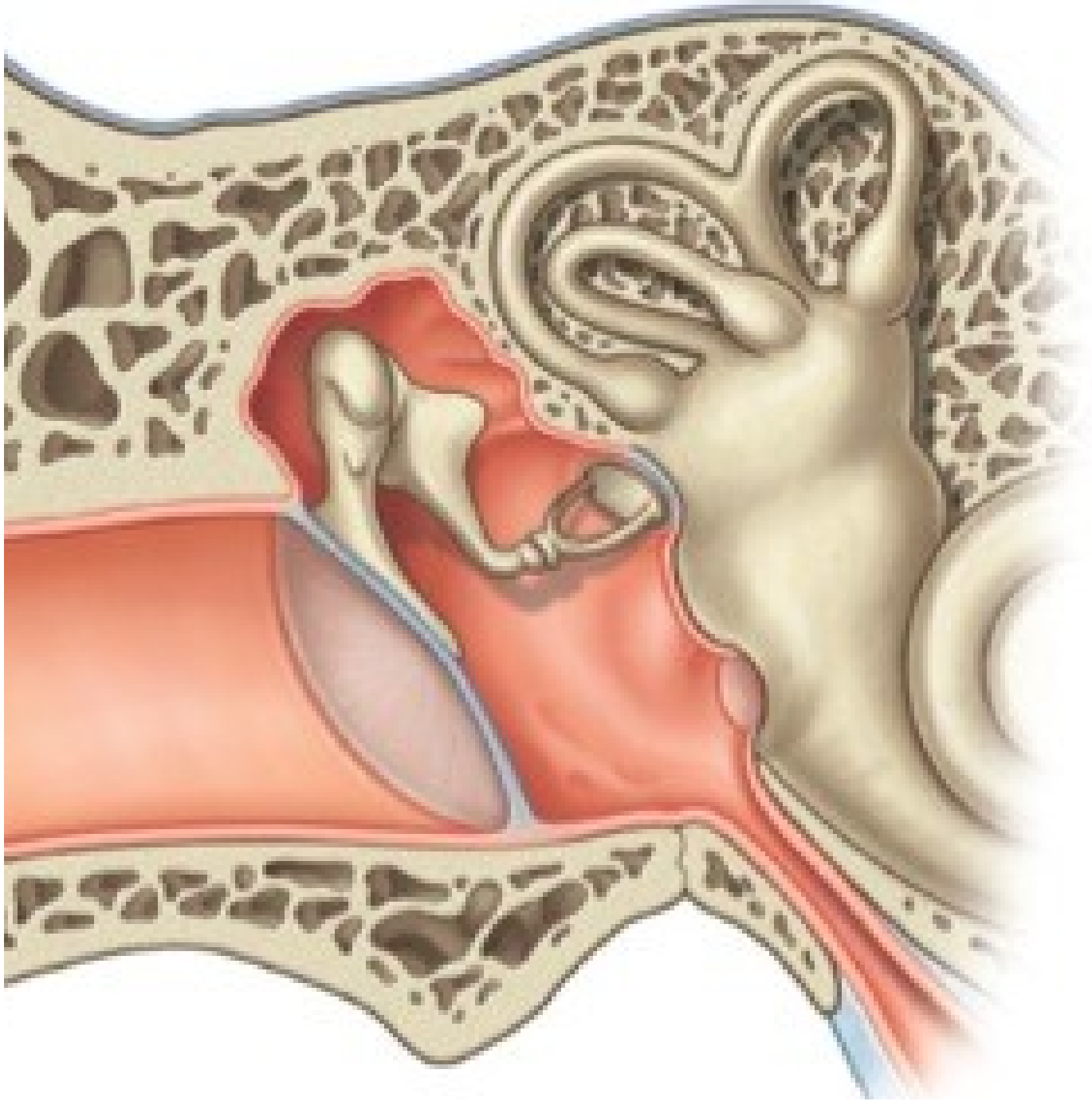
Auditory Ossicles

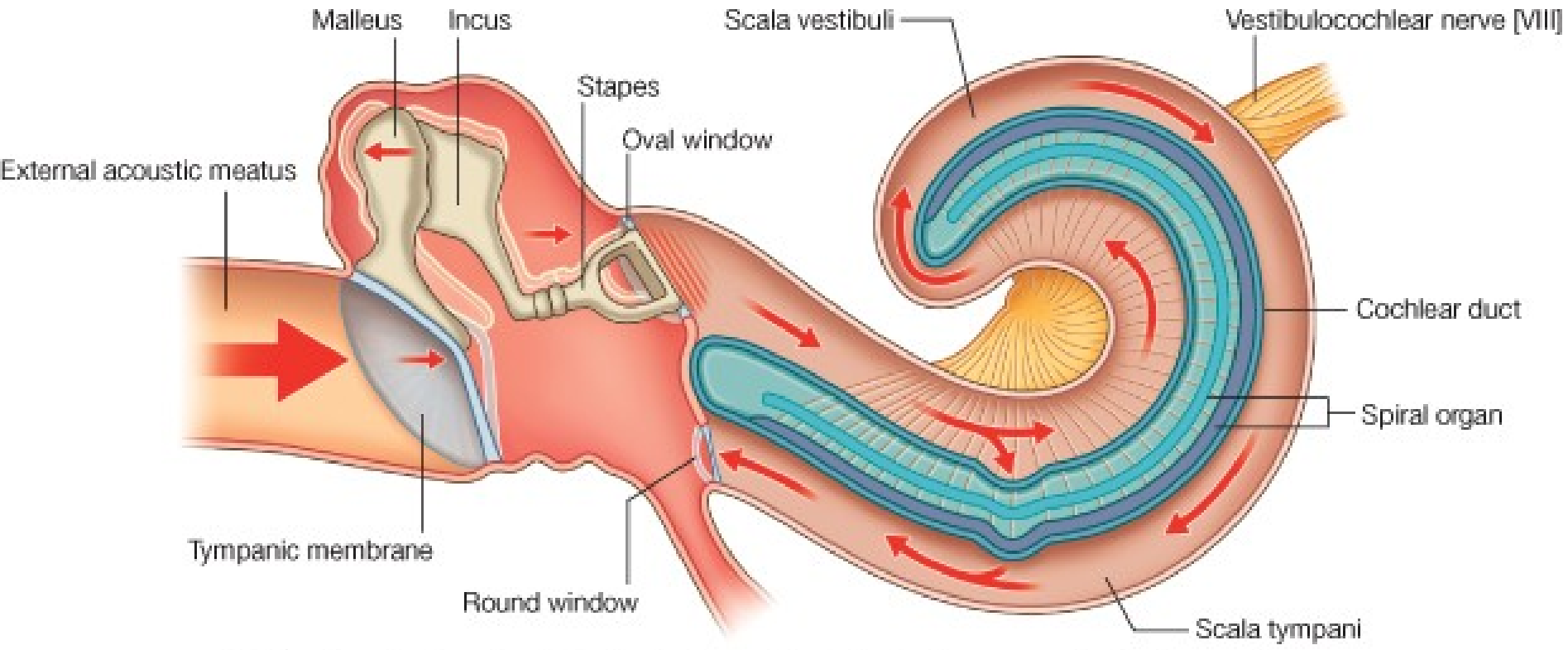
BONES

The bones of the middle ear are called the **auditory ossicles.** They are the malleus, incus and stapes. They are connected in a chain-like manner, linking the tympanic membrane to the **oval window** of the internal ear.
The **malleus** is the largest and most lateral of the ear bones, attaching to the tympanic membrane, via the **handle** of malleus. The **head** of the malleus lies in the **epitympanic recess**, where it articulates with the next auditory ossicle, the **incus.**
The next bone, the **incus**, consists of a **body** and two **limbs**. The body articulates with the malleus, the short limb attaches to the posterior wall of the middle, and the long limb joins the last of the ossicles; the **stapes**.
The **stapes** is the smallest bone in the human body. It joins the incus to the **oval window** of the inner ear. It is stirrup-shaped, with a **head**, two limbs, and a **base**. The head articulates with the incus, and the base joins the oval window.



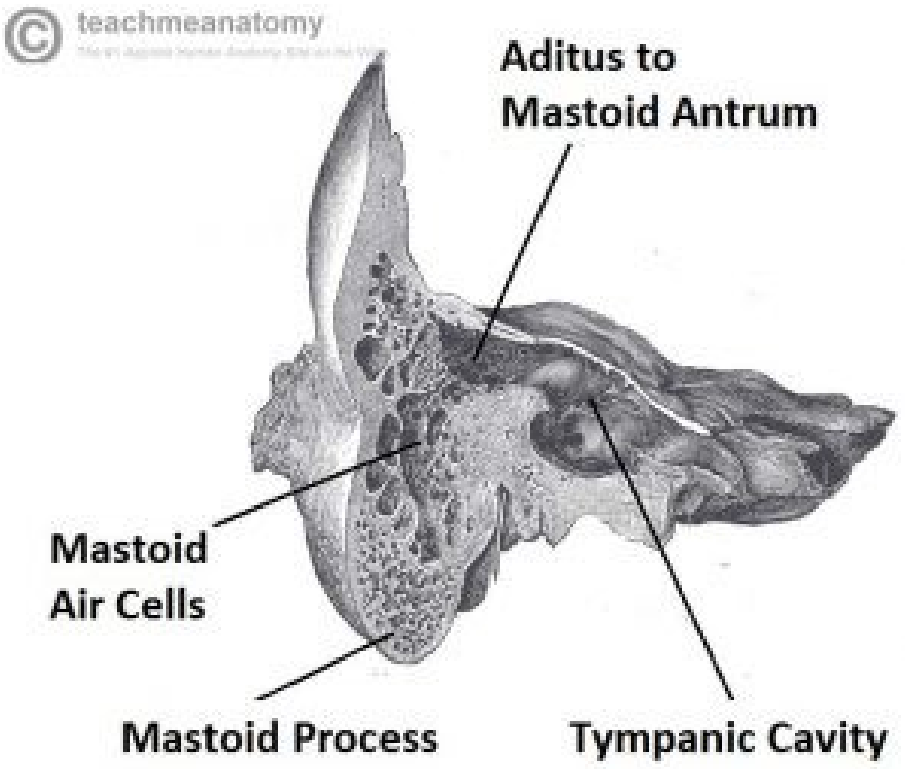

Lateral
Malleus
Head of malleus
Incudomallear joint
Body of incus
Short limb
Lateral process
Incus
Long limb
Anterior process
Incudostapedial joint
Handle of malleus
Posterior limb
Anterior limb
Base of stapes
Stapes
Anteri
or
Medial
Posteri
or




Malleus
Incus
Stapes
Oval window
External acoustic meatus
Tympanic membrane
Round window
Scala vestibuli
Vestibulocochlear nerve [VIII]
Cochlear duct
Spiral organ
Scala tympani

MASTOID AIR CELLS

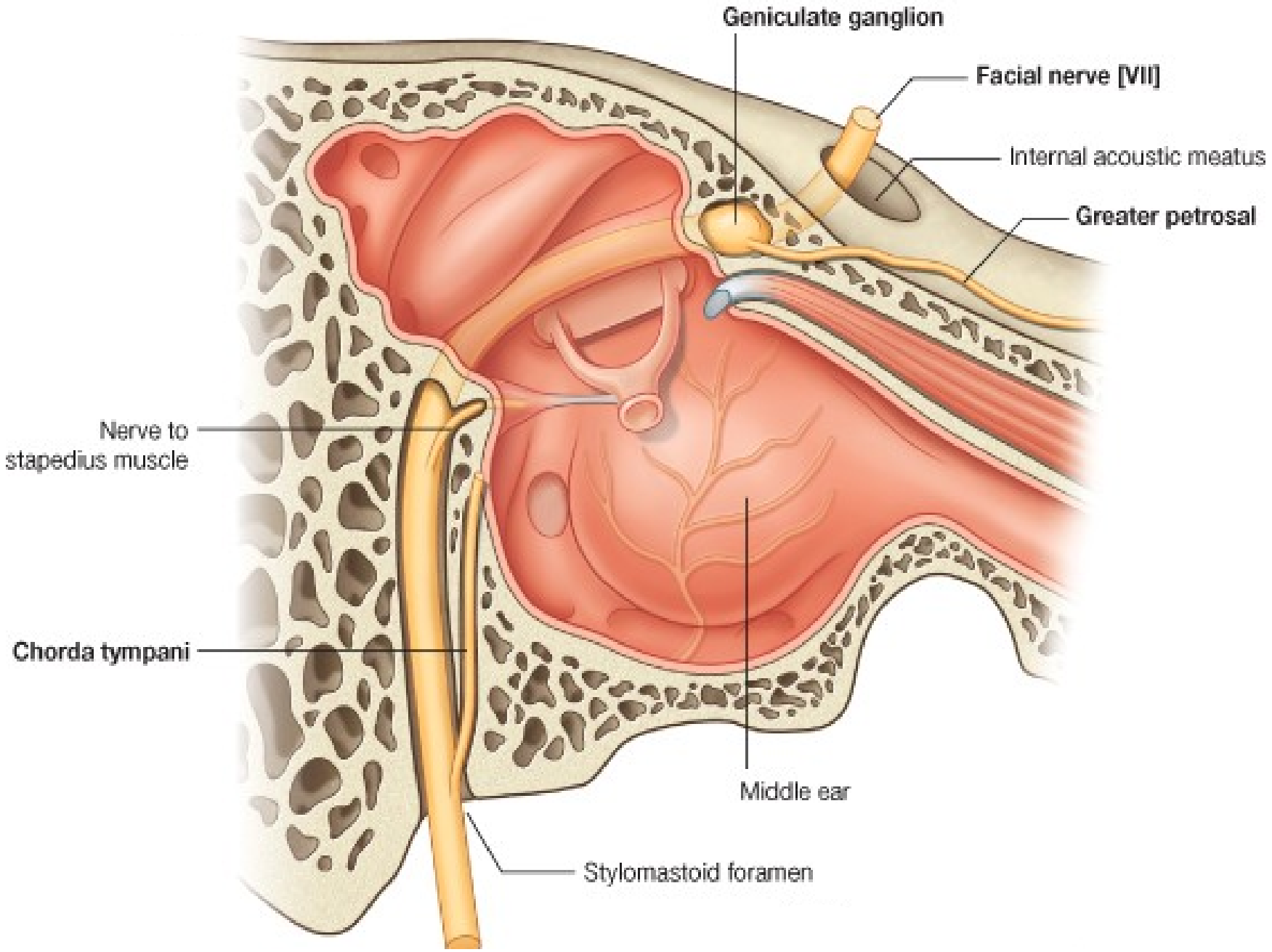
The mastoid air cells are located **posterior** to epitympanic recess. They are a collection of **air-filled** spaces in the mastoid process of the temporal bone. The air cells are contained within a cavity called the mastoid **antrum**. The mastoid antrum communicates with the middle ear via the **aditus** to mastoid antrum.
The mastoid air cells act as a '**buffer system**' of air – releasing air into the tympanic cavity when the pressure is too low.



Muscles of tympanic cavity

MUSCLES

There are two muscles which serve a **protective** function in the middle ear; the tensor tympani and stapedius. They contract in **response** to loud noise, inhibiting the vibrations of the malleus, incus and stapes, and reducing the transmission of sound to the inner ear. This action is known as the **acoustic reflex**.
The **tensor tympani** originates from the auditory tube and attaches to the handle of malleus, pulling it medially when contracting. It is innervated by a branch of the **mandibular** nerve. The **stapedius** muscle attaches to the stapes, and is innervated by the **facial** nerve.


Geniculate ganglion
Facial nerve [VII]
Internal acoustic meatus
Greater petrosal
Nerve to stapedius muscle
Chorda tympani
Middle ear
Stylomastoid foramen

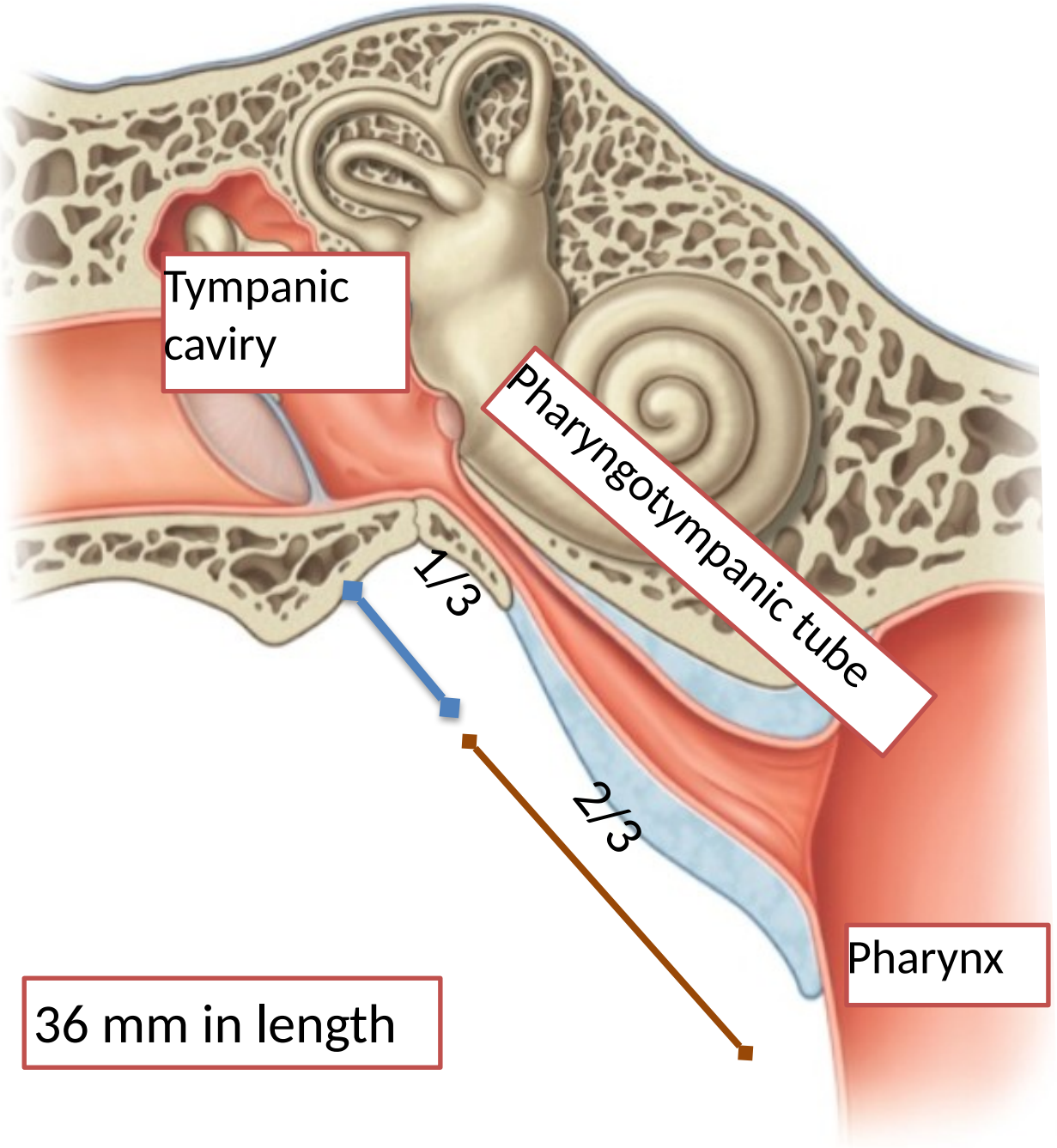
Action

- The tensor tympani and stapedius usually contract together in reflex response to sounds of fairly high intensity, exerting a '**protective damping effect** before vibrations reach the internal ear
- The tensor pulls the tympanic membrane inwards to tense it and also pushes the stapes more tightly into the fenestra vestibuli. The stapedius muscle opposes the tensor in the latter action.

Pharyngotympanic (Auditory) tube

AUDITORY TUBE

The auditory tube (eustachian tube) is a **cartilaginous** and **bony** tube that connects the middle ear to the **nasopharynx**. It acts to **equalise** the pressure of the middle ear to that of the external auditory meatus.
It extends from the anterior wall of the middle ear, in anterior, medioinferior direction, opening onto the **lateral** wall of the nasopharynx. In joining the two structures, it is a pathway by which an upper respiratory infection can spread into the middle ear.


Tympanic caviry
Pharyngotympanic tube
1/3
2/3
Pharynx
36 mm in length

The internal ear (Labyrinth)

The inner ear is the most distal part of the ear, housing the vestibulocochlear organs. It has two main functions:

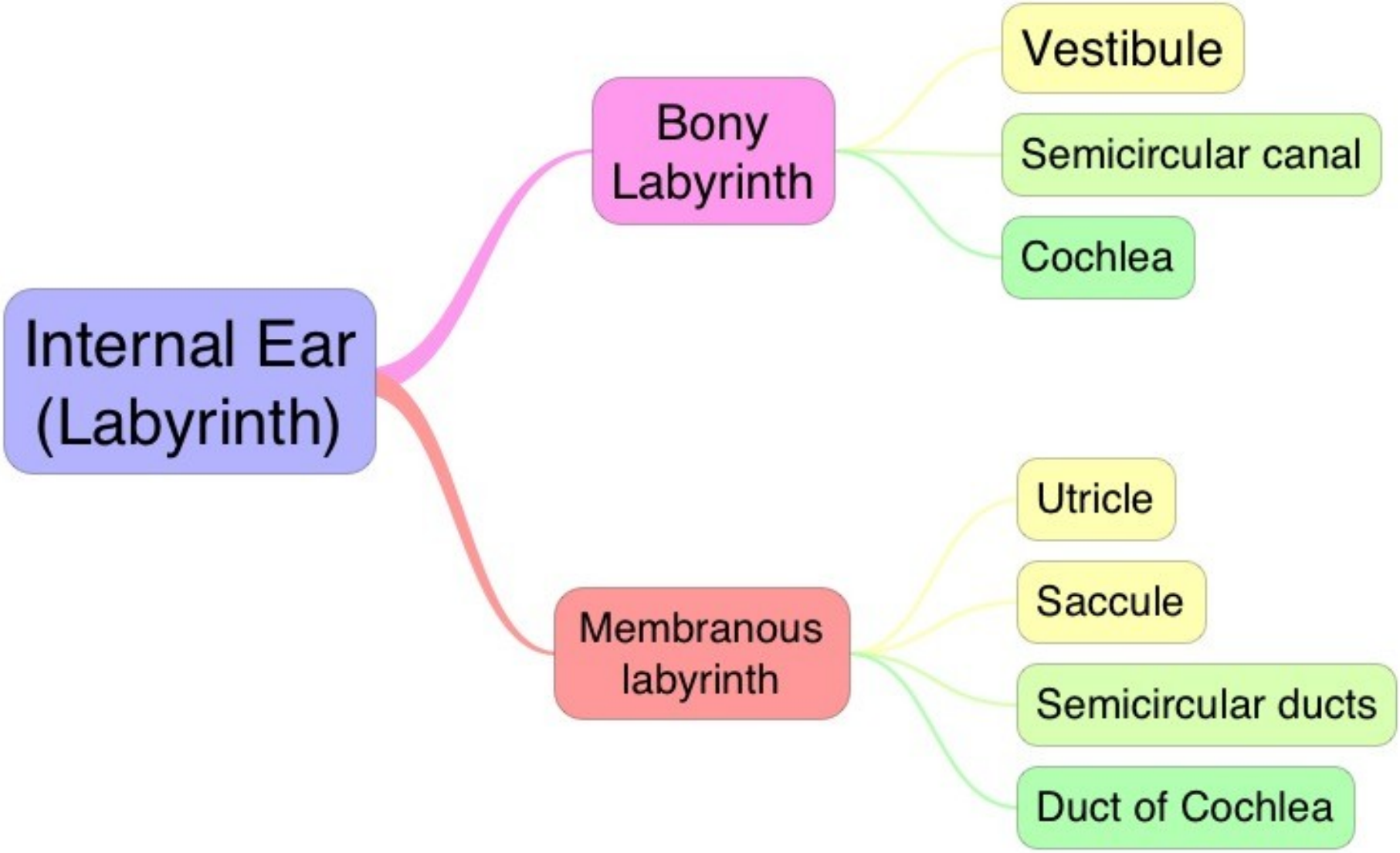
- To convert **mechanical** signals from the middle ear into **electrical** signals, which can transfer information to the auditory pathway in the brain.
- To maintain balance by **detecting position** and **motion**.

- The labyrinth is situated in the petrous part of the temporal bone, medial to the middle ear.
- It consists of the bony labyrinth, comprising a series of cavities within the bone.
- and the membranous labyrinth, comprising a series of membranous sacs and ducts contained within the bony labyrinth.

The inner ear is located within the **petrous** part of the **temporal bone**. It lies between the middle ear and the internal acoustic meatus, which lie laterally and medially respectively. The inner ear has two main components – the bony labyrinth and membranous labyrinth.

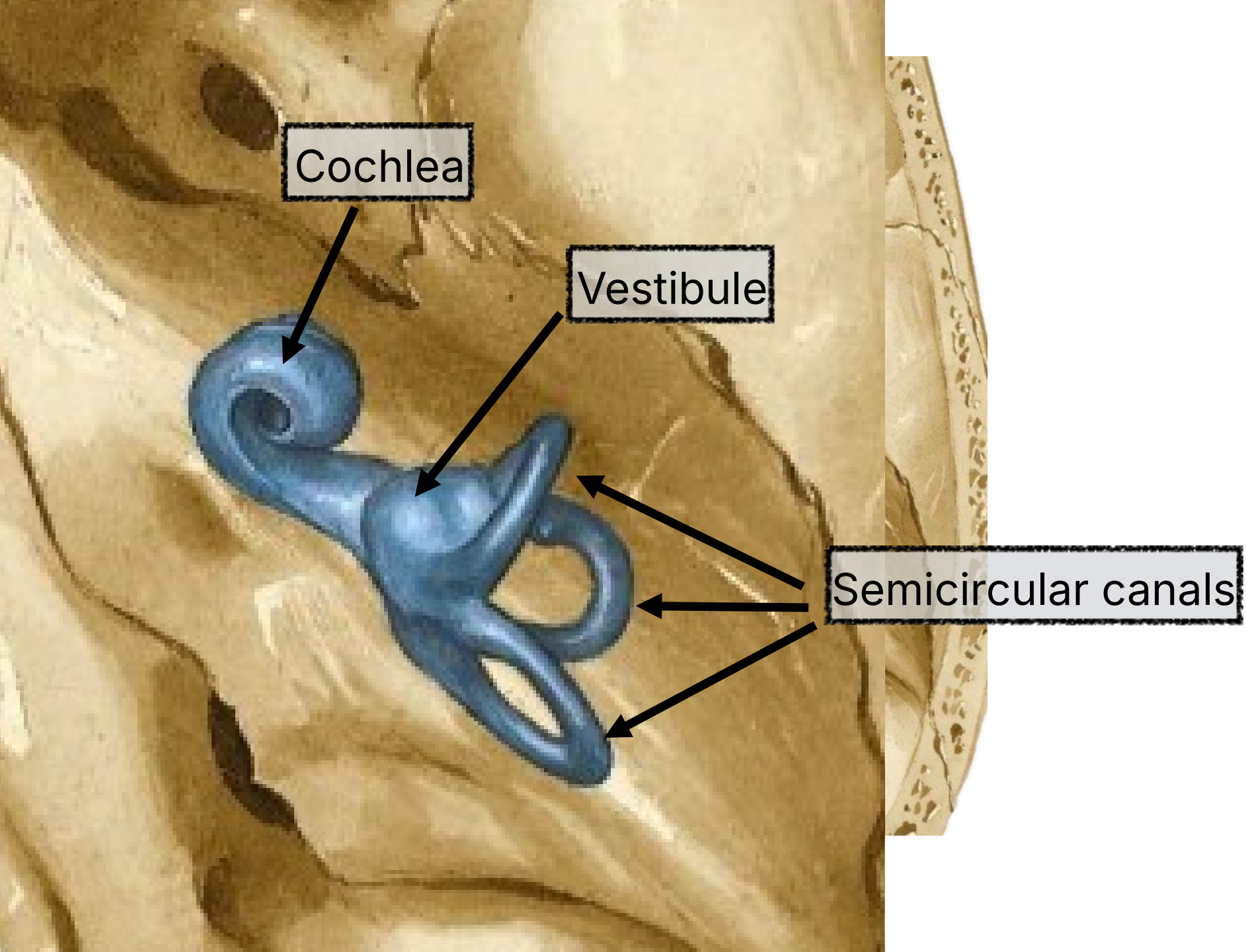
- **Bony labyrinth** – consists of a series of bony cavities within the petrous part of the temporal bone. It is composed of the cochlea, vestibule and three semicircular canals. All these structures are lined internally with periosteum and contain a fluid called perilymph.
- **Membranous labyrinth** – lies within the bony labyrinth. It consists of the cochlear duct, semicircular ducts, utricle and the saccule. The membranous labyrinth is filled with fluid called endolymph.

The inner ear has two openings into the middle ear, both covered by membranes. The **oval window** lies between the middle ear and the vestibule, whilst the **round window** separates the middle ear from the scala tympani (part of the cochlear duct).


Internal Ear (Labyrinth)
Bony Labyrinth
Vestibule
Semicircular canal
Cochlea
Membranous labyrinth
Utricle
Saccule
Semicircular ducts
Duct of Cochlea

Bony Labyrinth

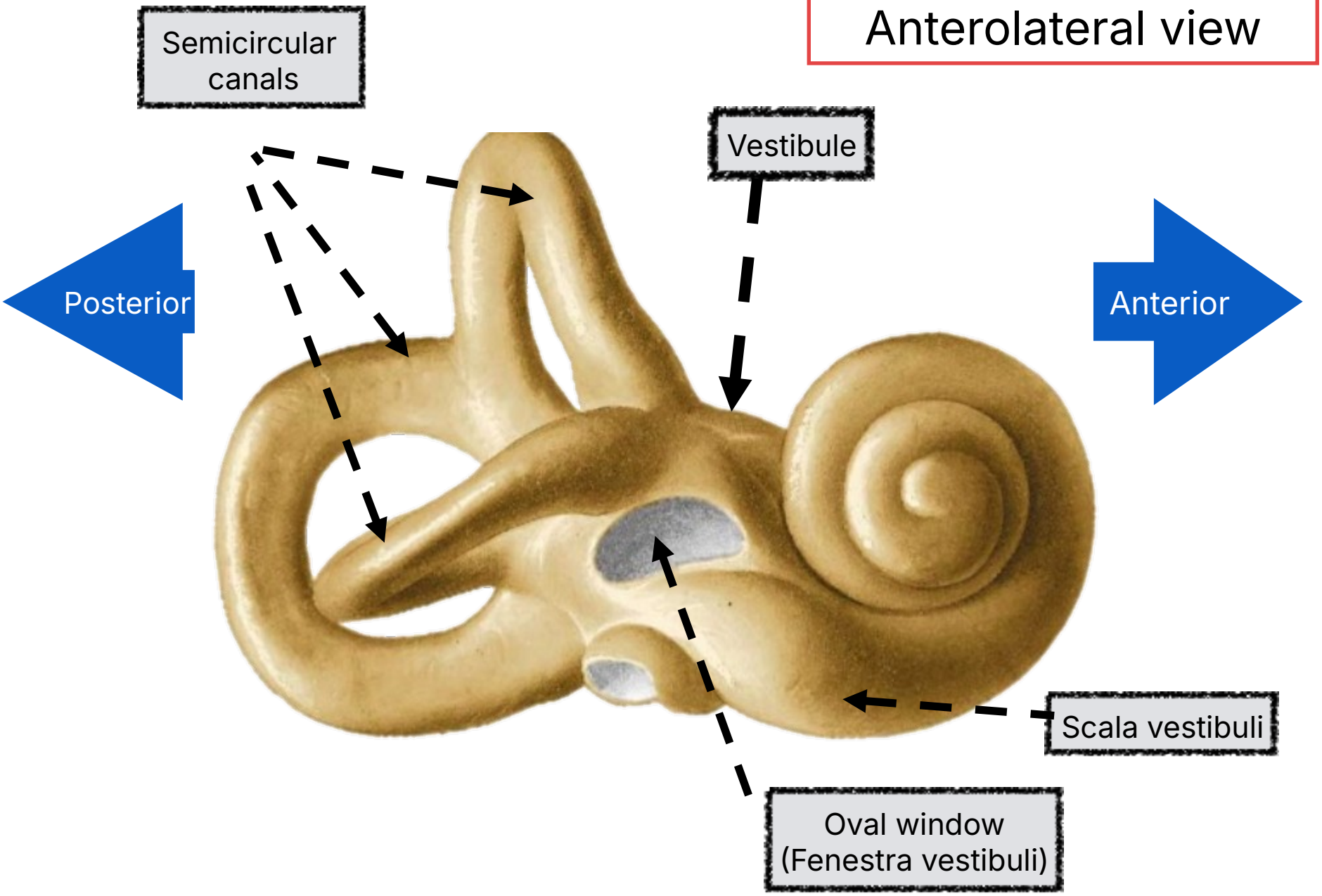
The bony labyrinth is a series of bony cavities within the petrous part of the temporal bone. It consists of three parts – the cochlea, vestibule and the three semicircular canals.


Cochlea
Vestibule
Semicircular canals

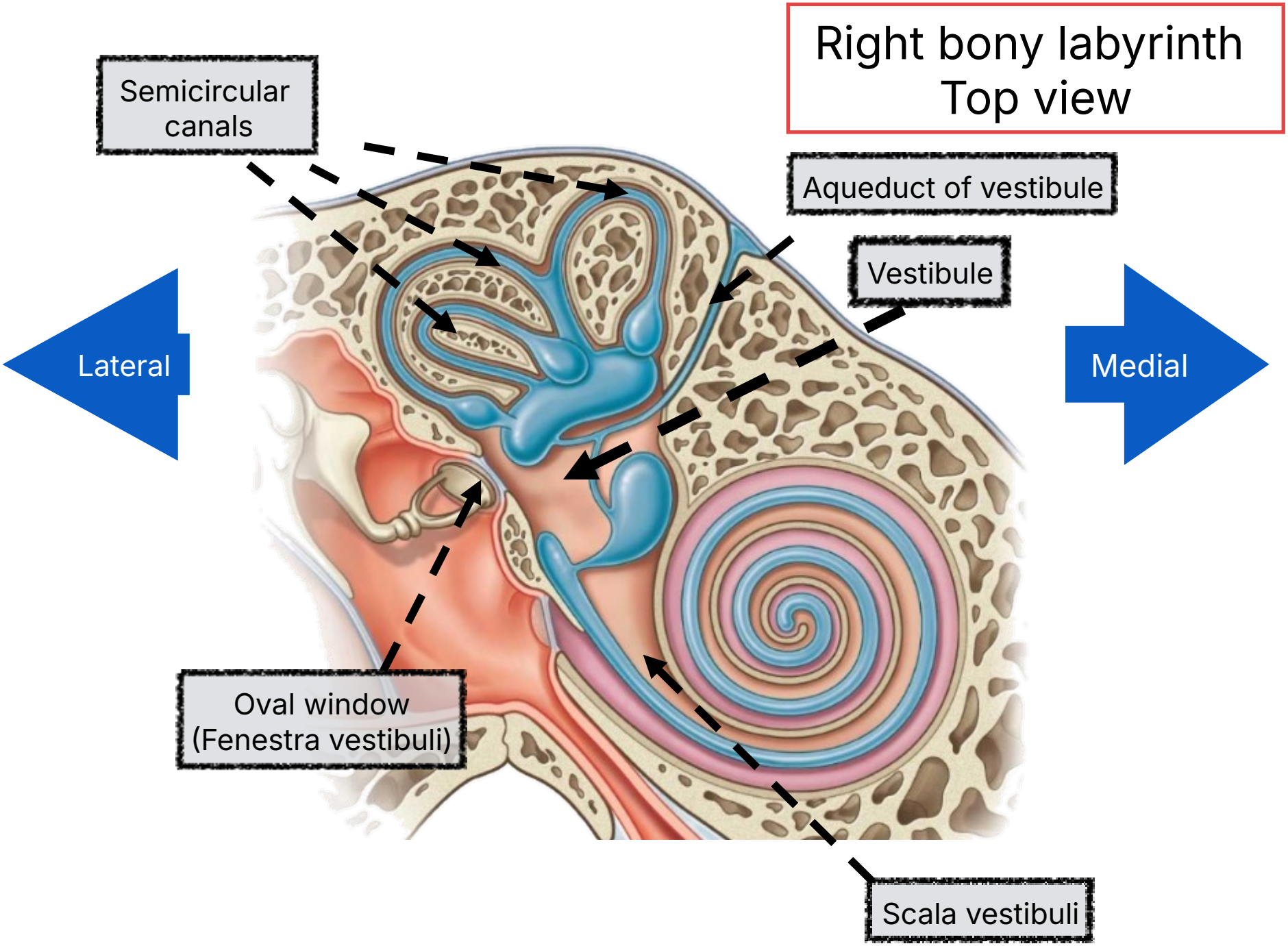
Vestibule

The vestibule is the central part of the bony labyrinth. It is separated from the middle ear by the **oval window**, and communicates anteriorly with the cochlea and posterioly with the semicircular canals. Two parts of the membranous labyrinth; the **saccule** and **utricle**, are located within the vestibule.

Right bony labyrinth
Anterolateral view



Right bony labyrinth
Top view



Cochlea

Cochlea
(snail shell -shaped)القوقعة

- It opens into the anterior part of the vestibule
- it consists of a central pillar, the modiolus, around which a hollow bony tube makes two and one half spiral turns
- Each successive turn is of decreasing radius so that the whole structure is conical. The apex faces anterolaterally and the base faces posteromedially
- The first basal turn of the cochlea is responsible for the promontory seen on the medial wall of the middle ear.

The cochlea houses the cochlea duct of the membranous labyrinth – the auditory part of the inner ear. It twists upon itself around a central portion of bone called the **modiolus,** producing a cone shape which points in an **anterolateral** direction. Branches from the cochlear portion of the **vestibulocochlear** (VIII) nerve are found at the base of the modiolus.

Extending outwards from the modiolus is a ledge of bone known as **spiral lamina,** which attaches to the cochlear duct, holding it in position. The presence of the cochlear duct creates two perilymph-filled chambers above and below:

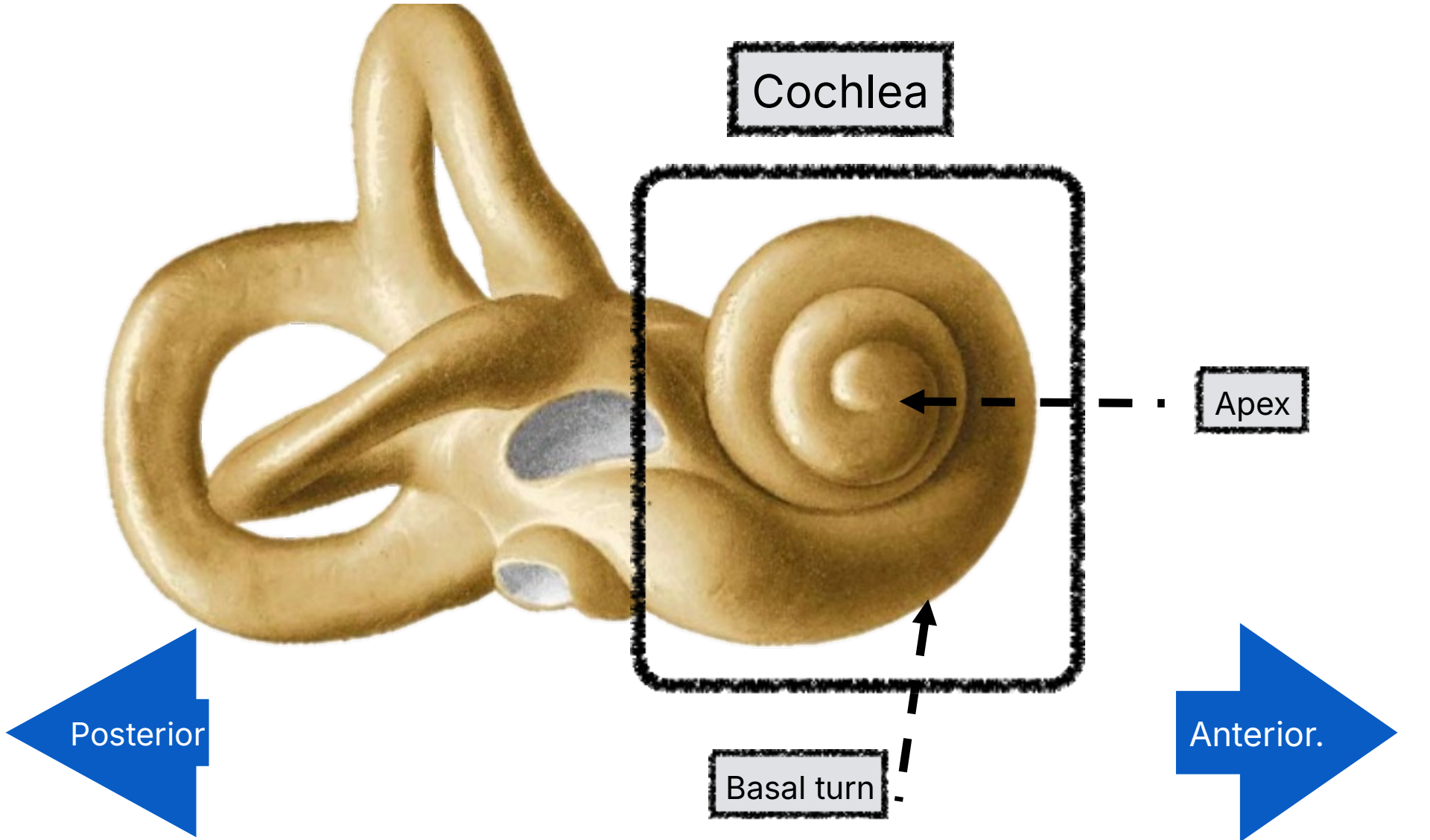
- **Scala vestibuli**: Located superiorly to the cochlear duct. As its name suggests, it is continuous with the vestibule.
- **Scala tympani**: Located inferiorly to the cochlear duct. It terminates at the round window.

Semicircular Canals

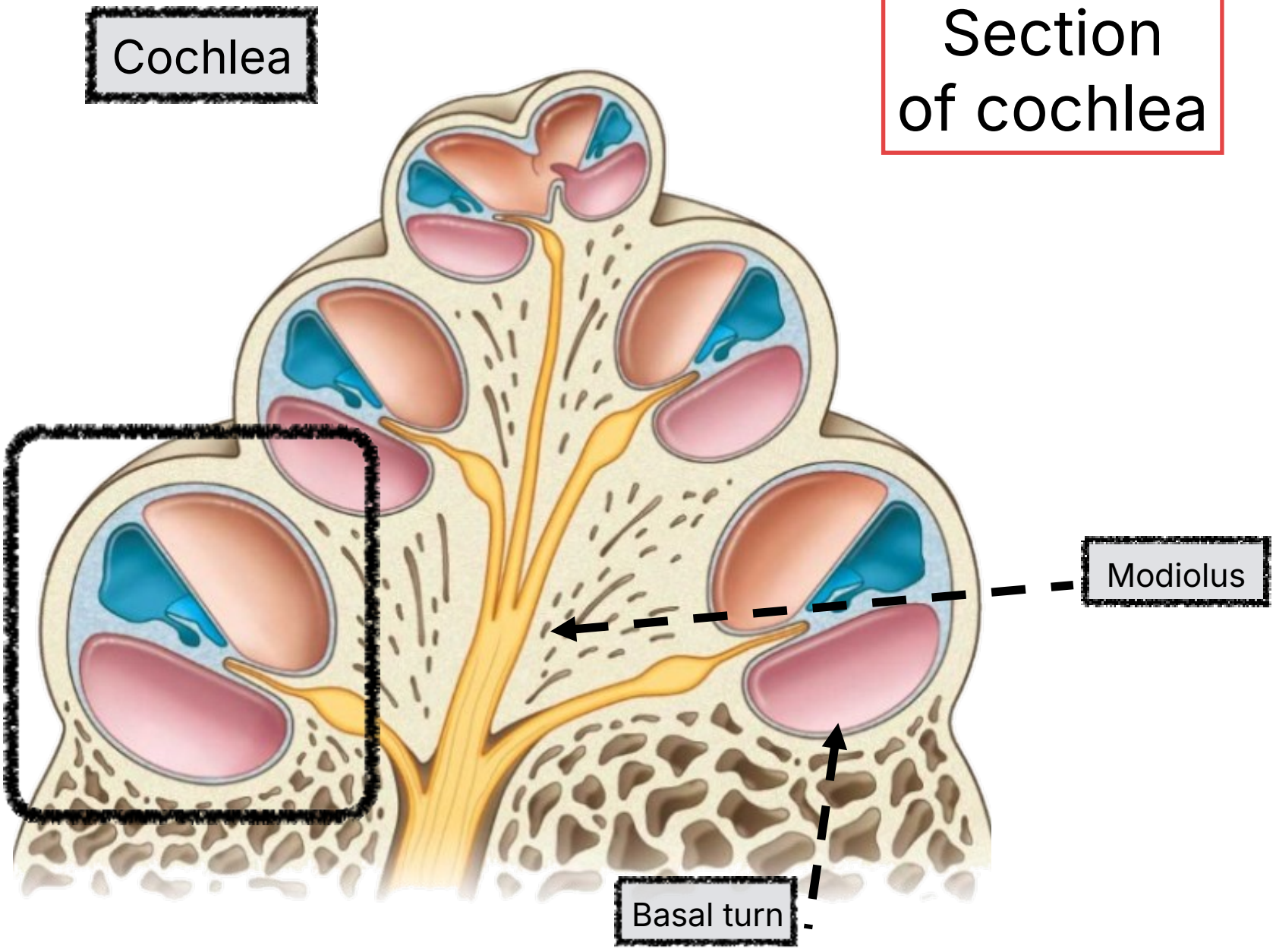
There are three semicircular canals; anterior, lateral and posterior. They contain the **semicircular ducts**, which are responsible for balance (along with the utricle and saccule).

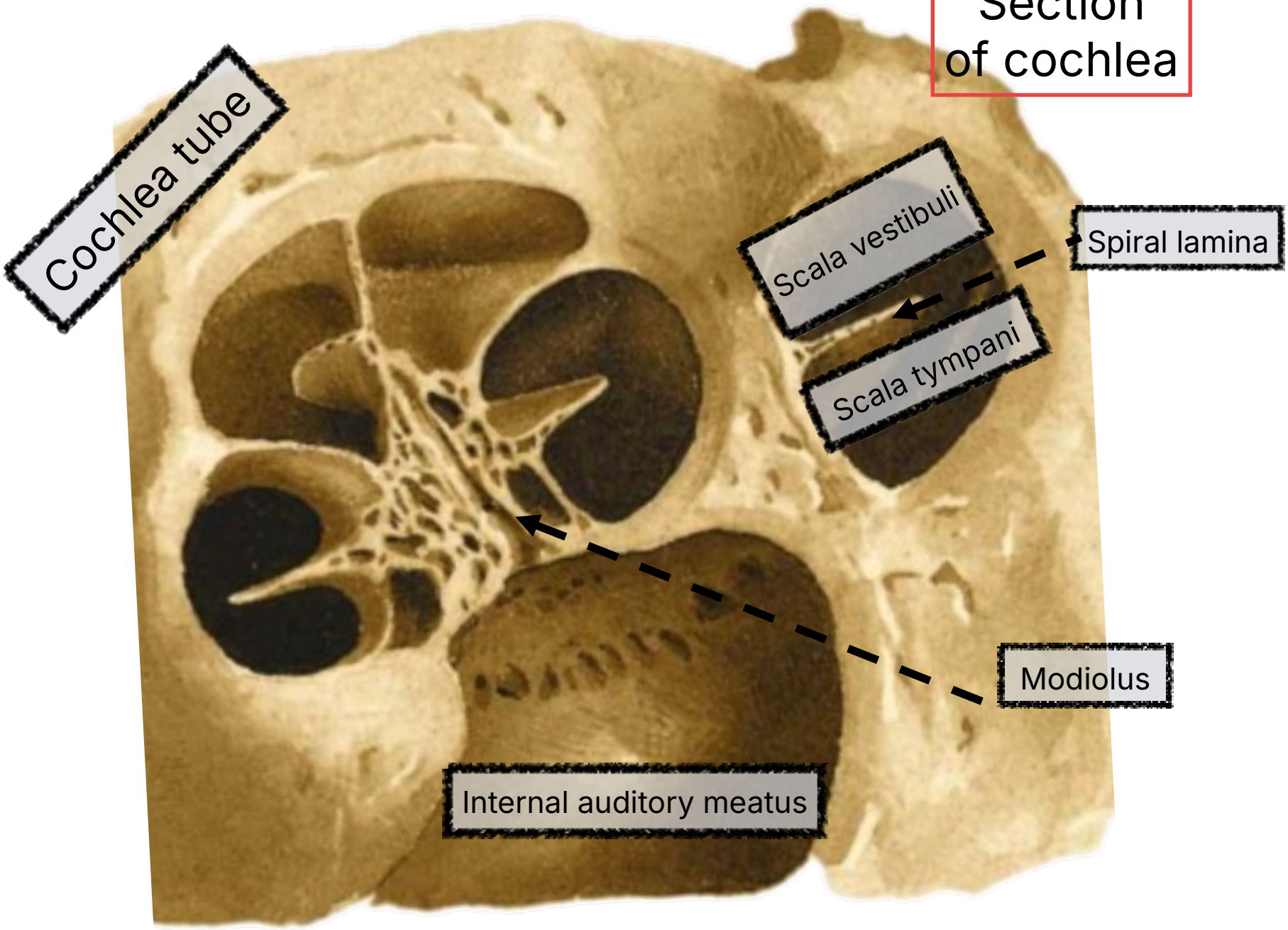
The canals are situated superoposterior to the vestibule, at right angles to each other. They have a swelling at one end, known as the **ampulla**.

Right bony labyrinth
Anterolateral view

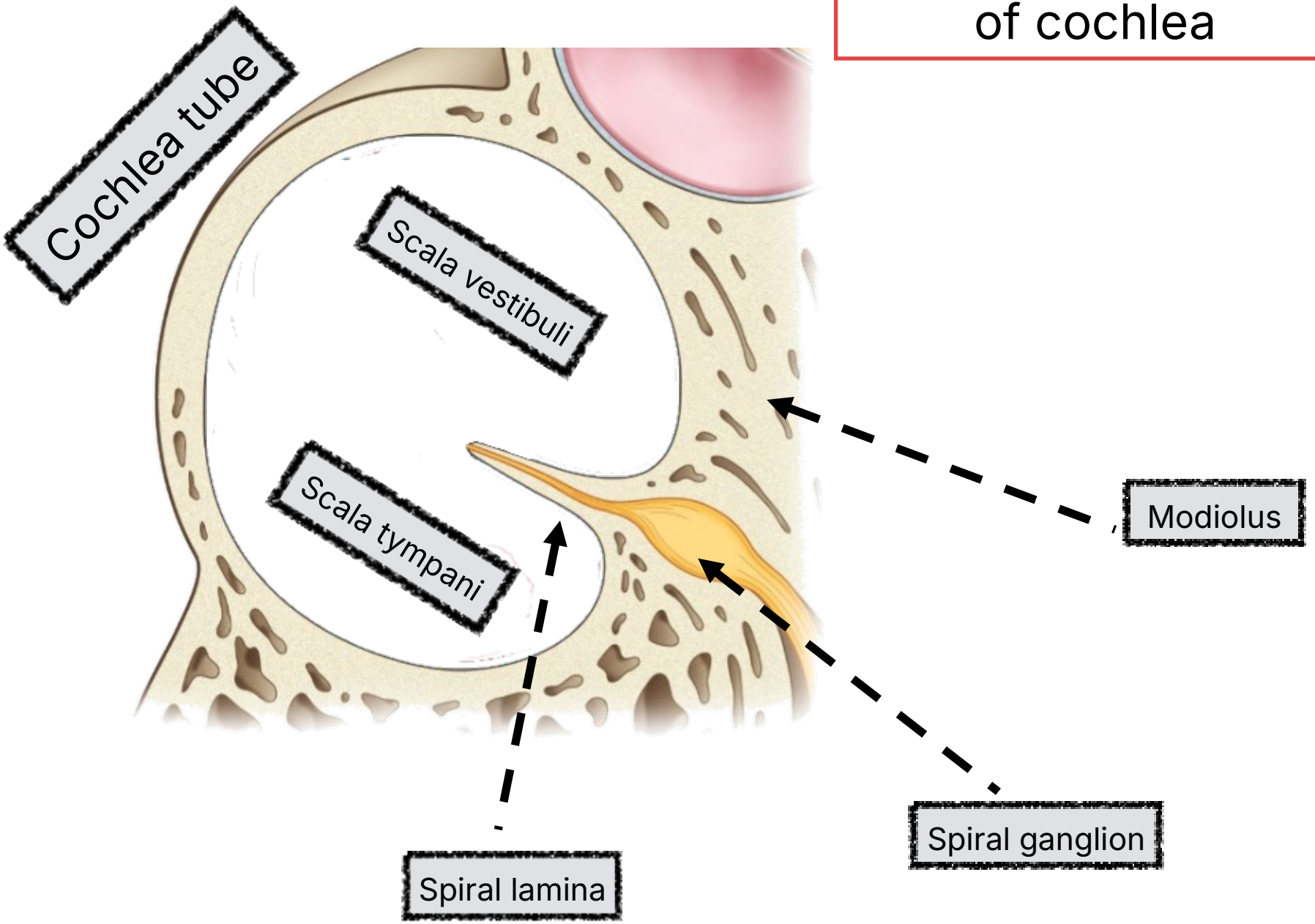


Section of cochlea




Section of cochlea
Cochlea tube
Scala vestibuli
Spiral lamina
Scala tympani
Modiolus
Internal auditory meatus

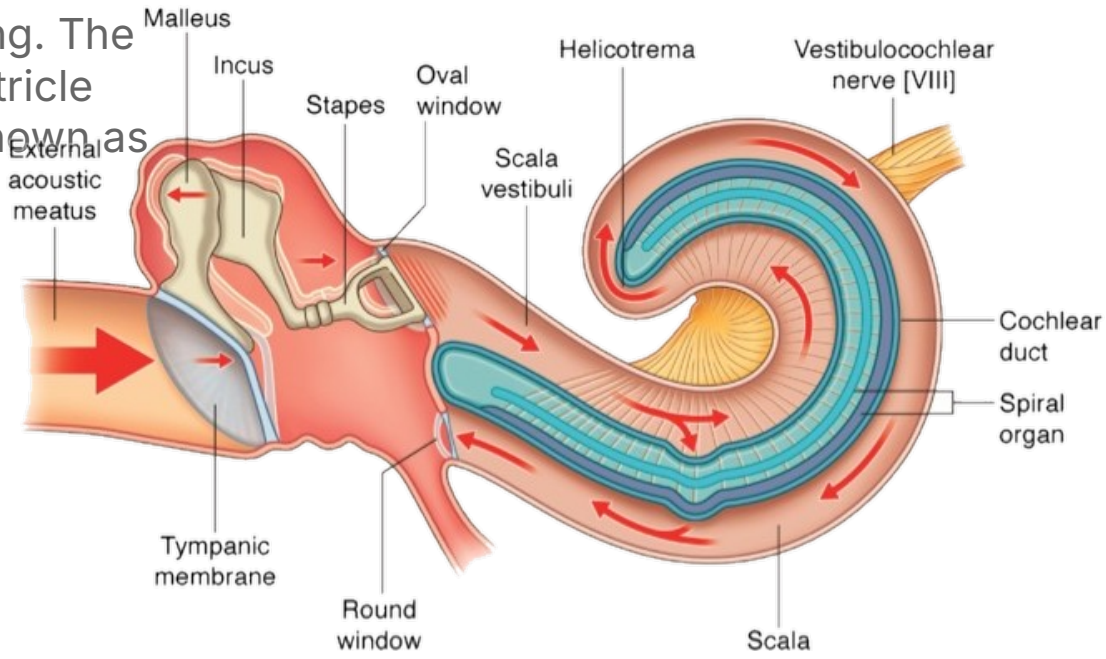
Longitudinal section of cochlea

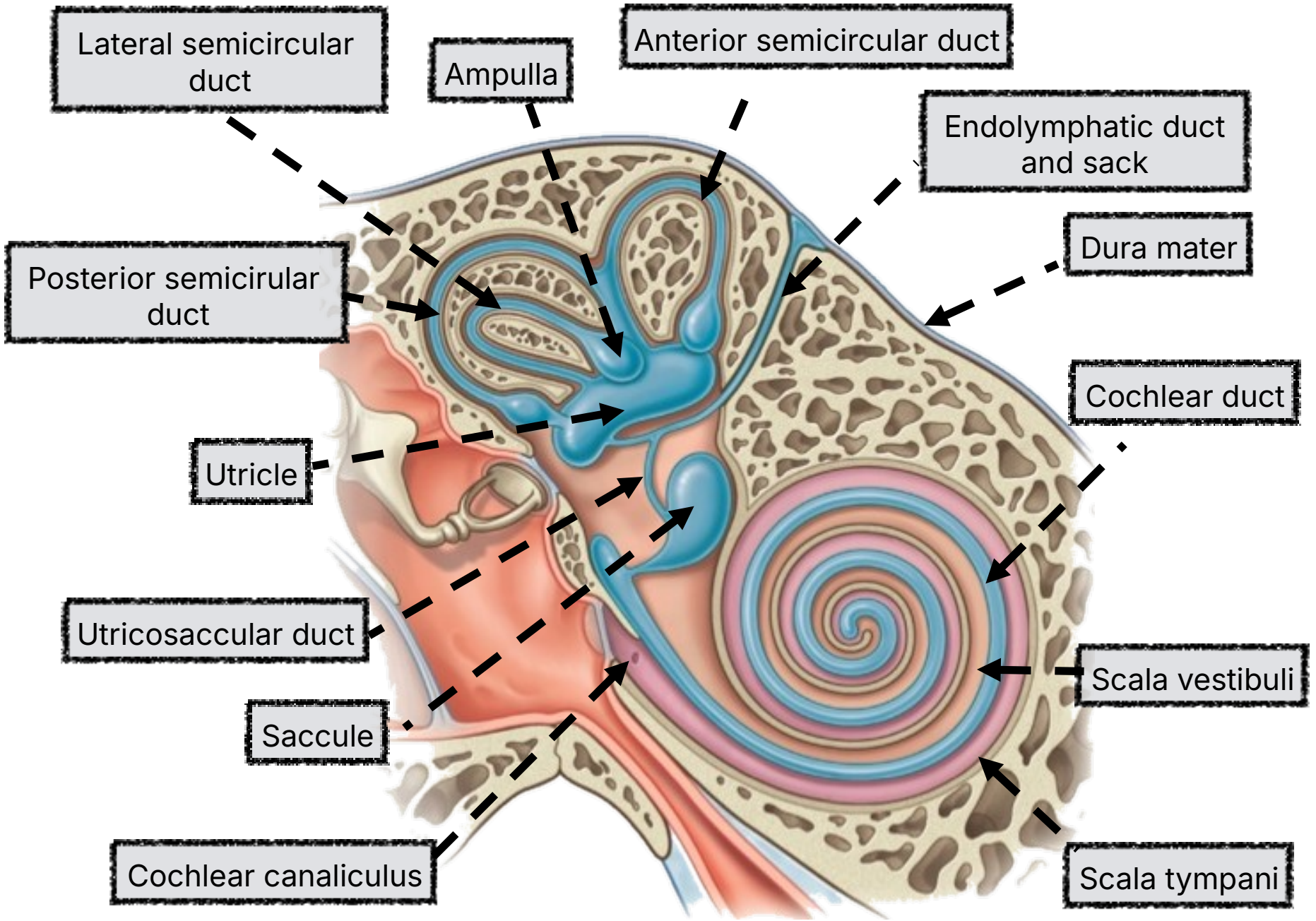


Membranous Labyrinth

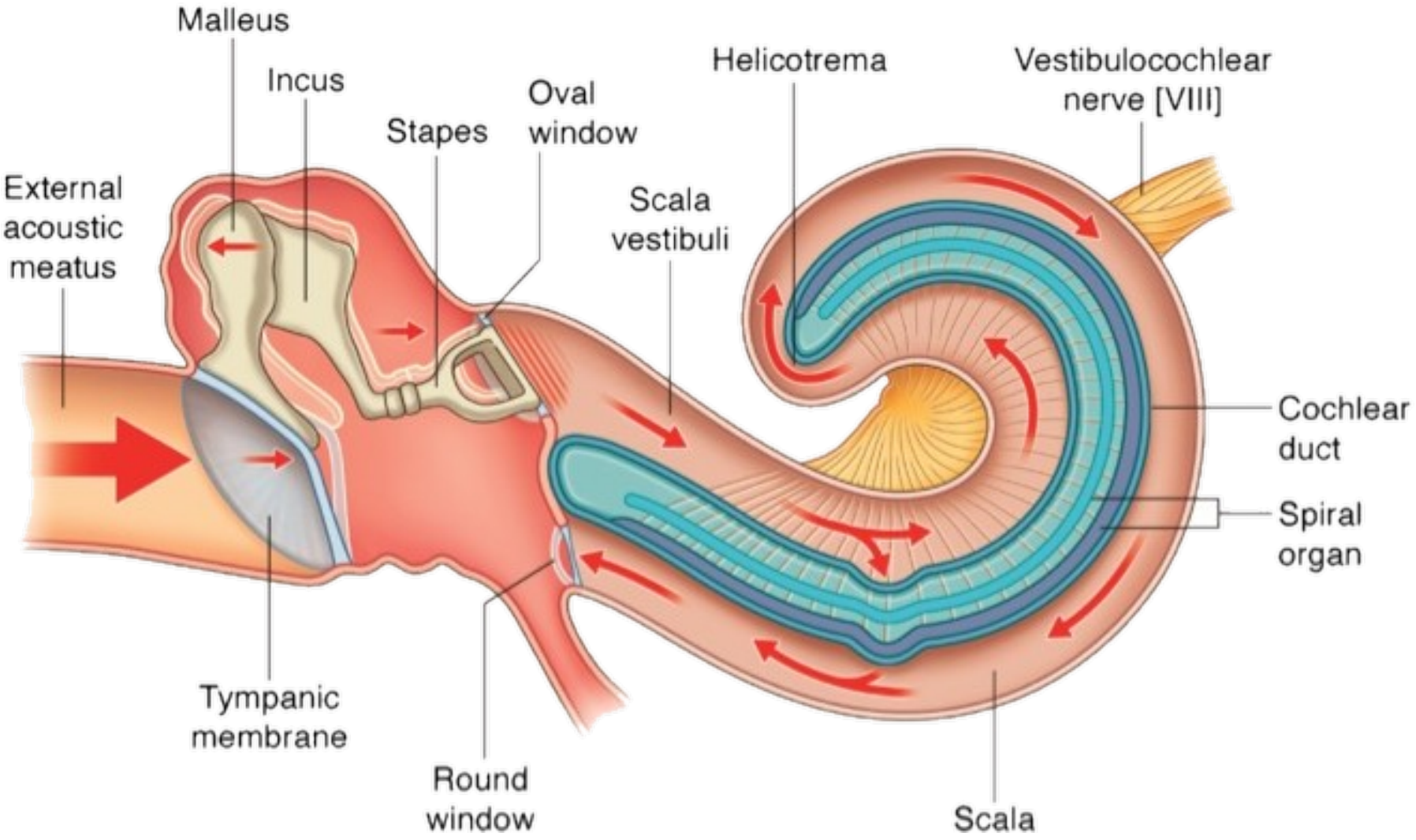
- it is contained within the bony labyrinth but much smaller in size.
- It is filled with endolymph
- Its walls receive the branches of the vestibulocochlear nerve.

The membranous labyrinth is a continuous system of ducts filled with **endolymph.** It lies within the bony labyrinth, surrounded by perilymph. It is composed of the cochlear duct, three semicircular ducts, saccule and the utricle.
The cochlear duct is situated within the cochlea and is the organ of hearing. The semicircular ducts, saccule and utricle are the organs of balance (also known as the **vestibular apparatus**).




Lateral semicircular duct
Ampulla
Anterior semicircular duct
Endolymphatic duct and sack
Dura mater
Posterior semicirular duct
Cochlear duct
Utricle
Utricosaccular duct
Scala vestibuli
Saccule
Cochlear canaliculus
Scala tympani

Labyrinth

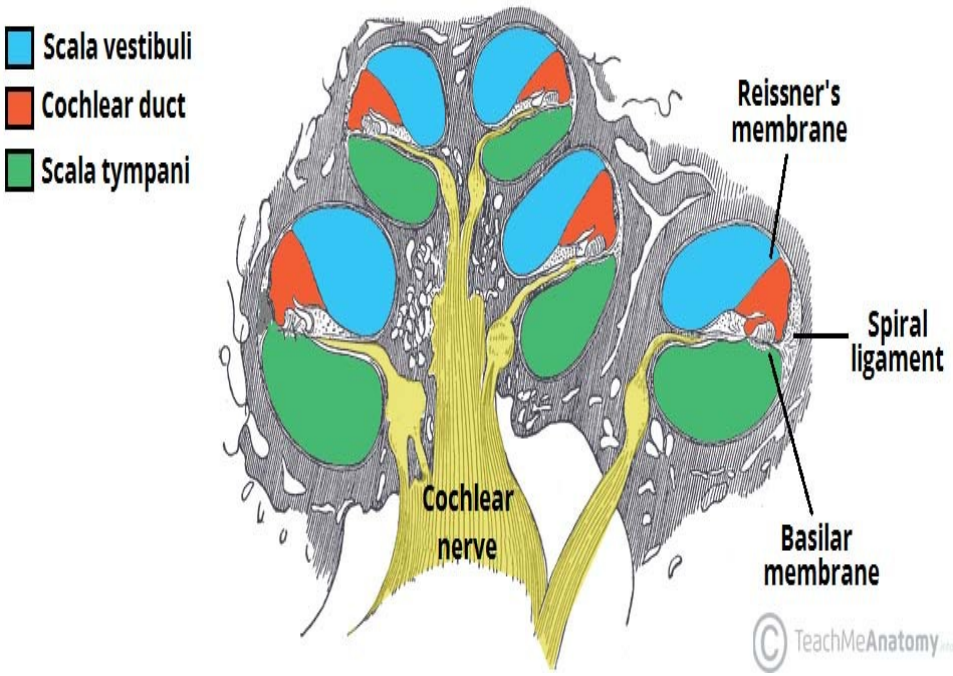

Malleus
Incus
Stapes
Oval window
Helicotrema
Vestibulocochlear nerve [VIII]
External acoustic meatus
Scala vestibuli
Cochlear duct
Spiral organ
Tympanic membrane
Round window
Scala

COCHLEAR DUCT

The cochlear duct is located within the bony scaffolding of the cochlea. It is held in place by the spiral lamina. The presence of the duct creates two canals above and below it – the **scala vestibuli** and **scala tympani** respectively. The cochlear duct can be described as having a triangular shape:

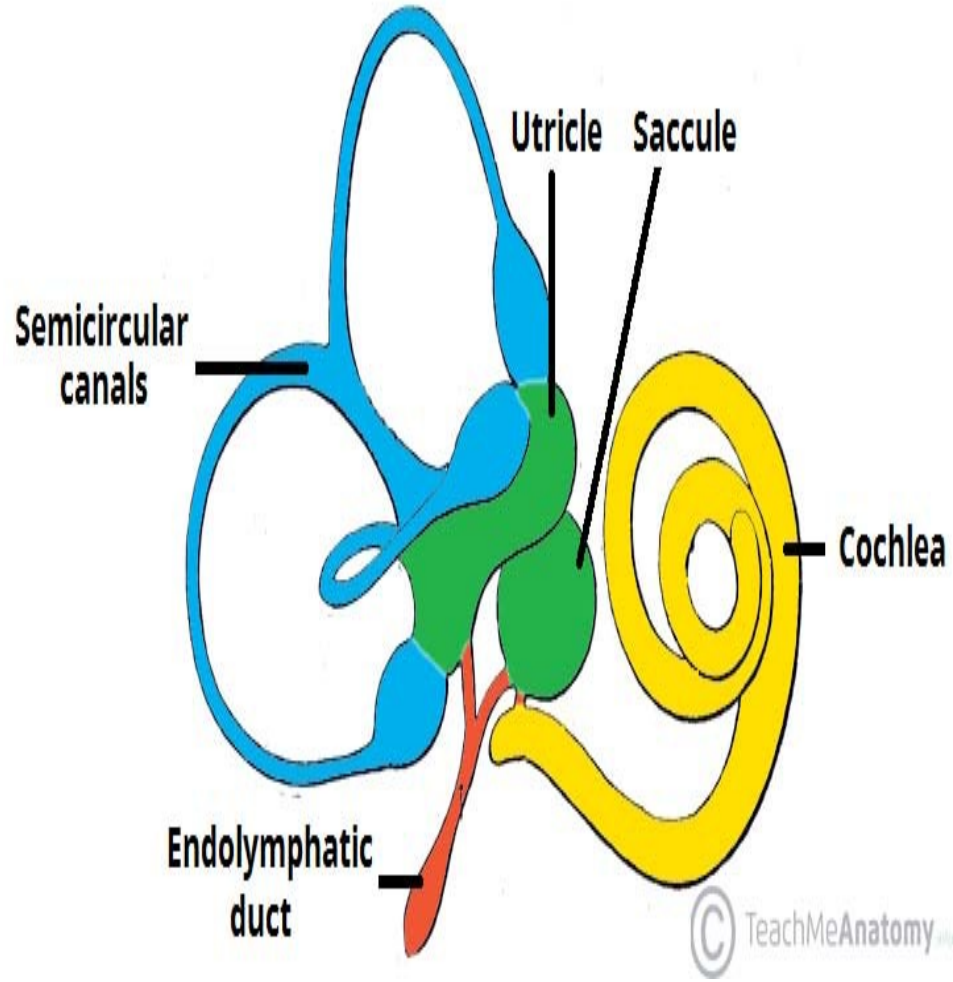
- **Lateral wall** – Formed by thickened periosteum, known as the spiral ligament.
- **Roof** – Formed by a membrane which separates the cochlear duct from the scala vestibuli, known as the Reissner's membrane.
- **Floor** – Formed by a membrane which separates the cochlear duct from the scala tympani, known as the basilar membrane.

The basilar membrane houses the epithelial cells of hearing – the **Organ of Corti.** A more detailed description of the Organ of Corti is beyond the scope of this article.



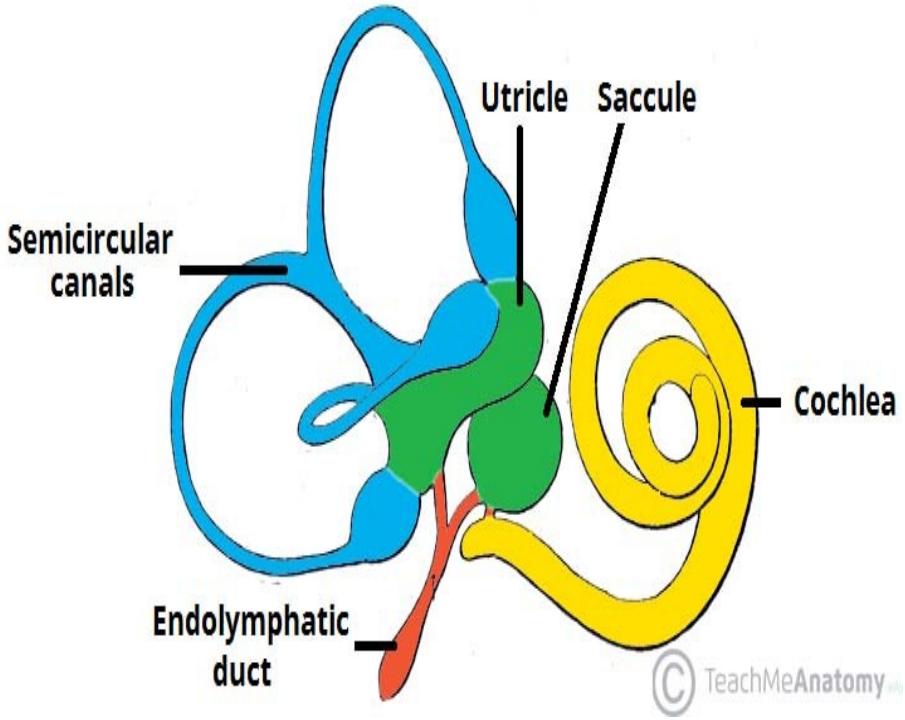
SACCULE AND UTRICLE

The saccule and utricle are two membranous sacs located in the vestibule. The **utricle** is the larger of the two, receiving the three semicircular ducts. The **saccule** is globular in shape and receives the cochlear duct. Endolymph drains from the saccule and utricle into the **endolymphatic duct**. The duct travels through the **vestibular aqueduct** to the posterior aspect of the petrous part of the temporal bone. Here, the duct expands to a sac where endolymph can be secreted and absorbed.

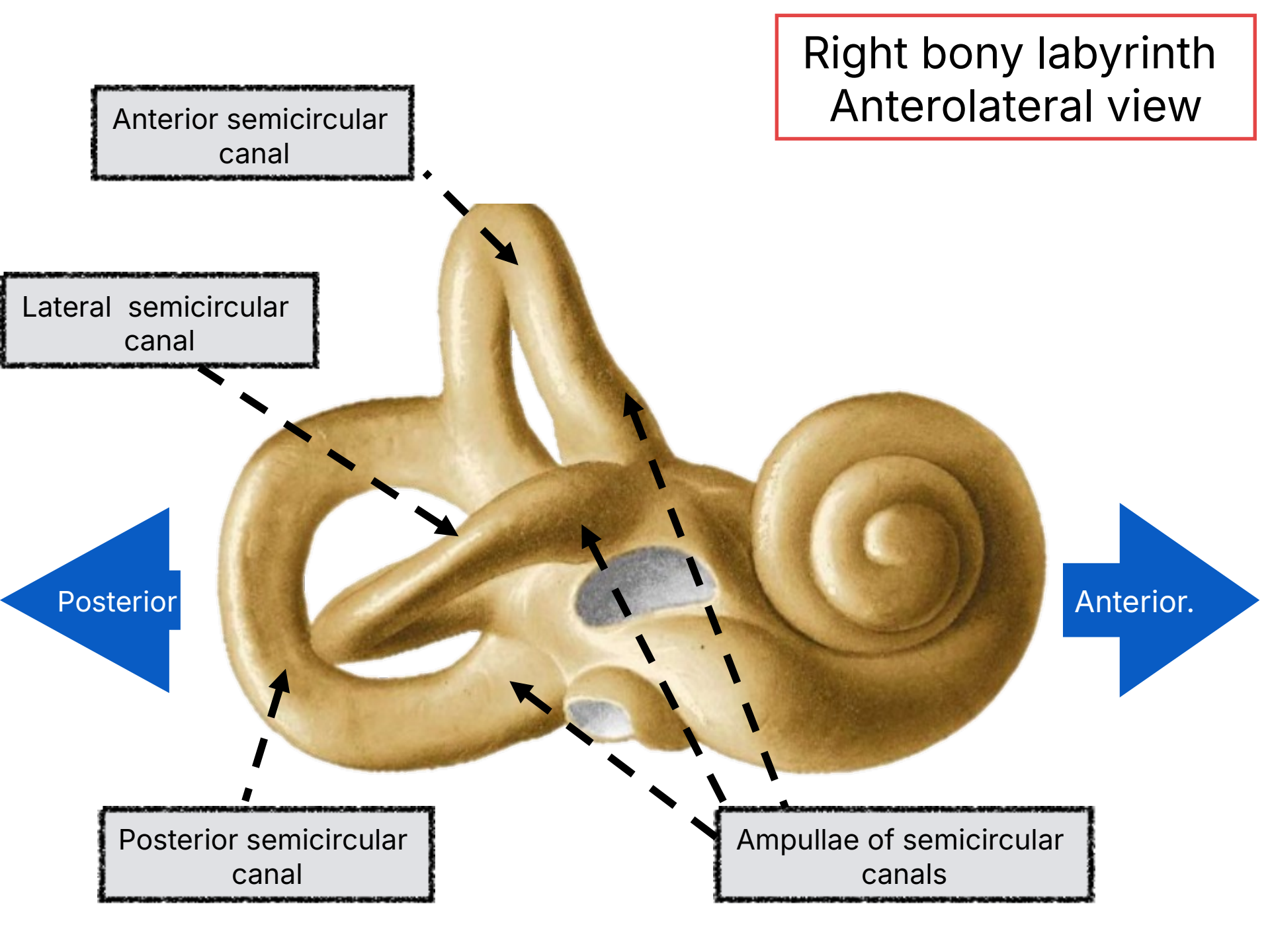


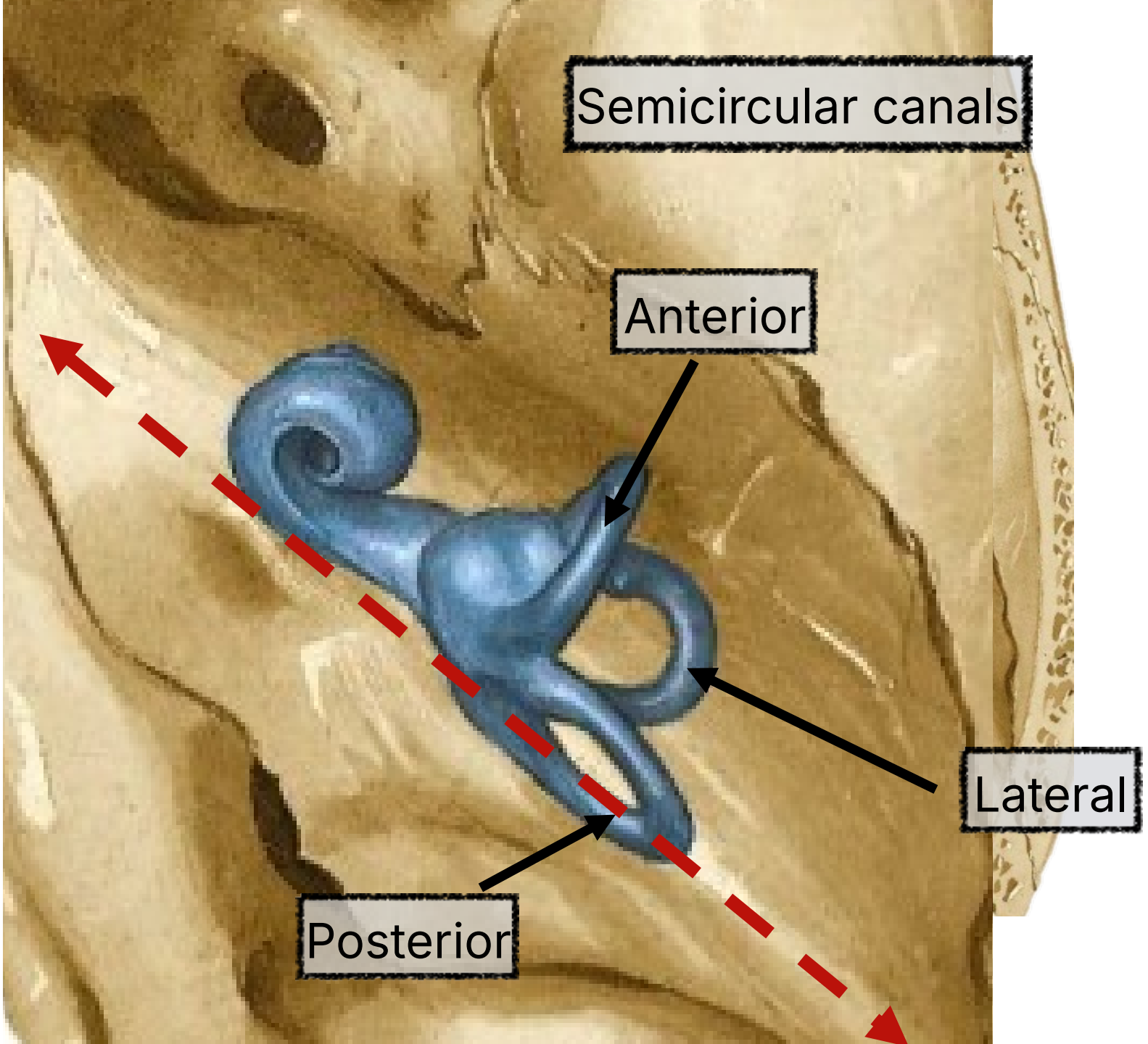
Semicircular canals

The semicircular ducts are located within the semicircular canals, and share their orientation. Upon movement of the head, the flow of **endolymph** within the ducts changes speed and/or direction. Sensory receptors in the ampullae of the semicircular canals detect this change, and send signals to the brain, allowing for the processing of balance.



Right bony labyrinth
Anterolateral view




Semicircular canals
Anterior
Lateral
Posterior

VASCULATURE

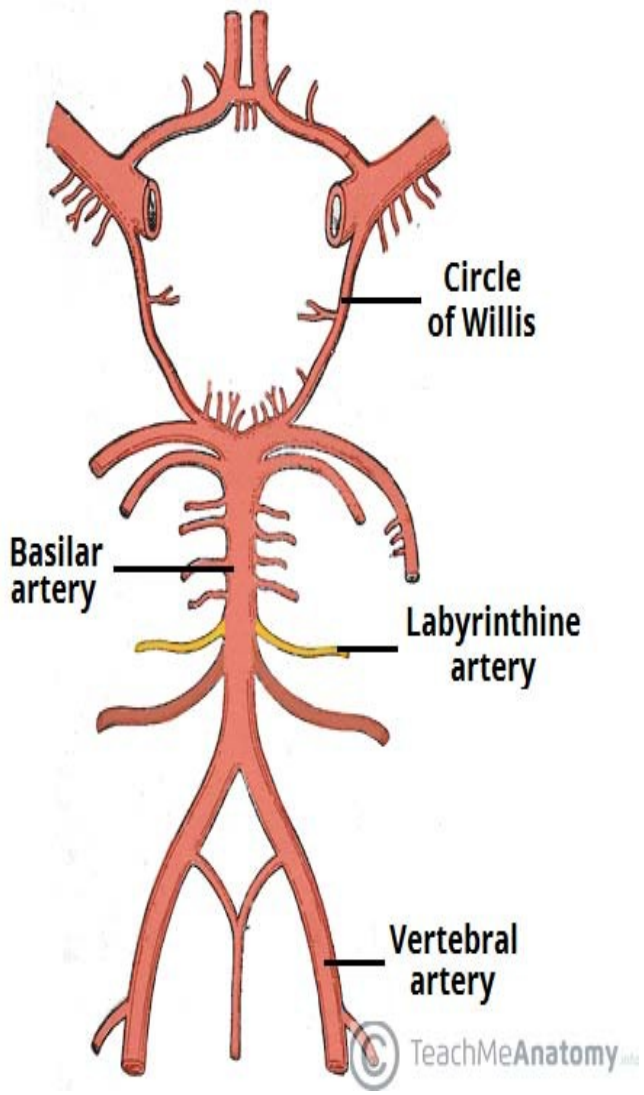
The bony labyrinth and membranous labyrinth have different arterial supplies. The bony labyrinth receives three arteries, which also supply the surrounding temporal bone:

- **Anterior tympanic branch** (from maxillary artery).
- **Petrosal branch** (from middle meningeal artery).
- **Stylomastoid branch** (from posterior auricular artery).

The membranous labyrinth is supplied by the **labyrinthine artery**, a branch of the inferior cerebellar artery (or, occasionally, the basilar artery). It divides into three branches:

- **Cochlear branch** – supplies the cochlear duct.
- **Vestibular branches (x2)** – supply the vestibular apparatus.

Venous drainage of the inner ear is through the **labyrinthine vein**, which empties into the sigmoid sinus or inferior petrosal sinus.



INNERVATION

The inner ear is innervated by the vestibulocochlear nerve (CN VIII). It enters the inner ear via the internal acoustic meatus, where it divides into the **vestibular nerve** (responsible for balance) and the **cochlear nerve** (responsible for hearing):

- **Vestibular nerve** – enlarges to form the vestibular ganglion, which then splits into superior and inferior parts to supply the utricle, saccule and three semicircular duct.
- **Cochlear nerve** – enters at the base of the modiolus and its branches pass through the lamina to supply the receptors of the Organ of Corti.

The facial nerve, CN VII, also passes through the inner ear, but does not innervate any of the structures present.

Thank You